

VLISM

Variable-Length Subnet Mask

Workbook

Version 2.0

Instructor's Edition

192.168.

192.168.10.96

192.168.10.126

172.31.15.0

10.250.1.0

IP Address Classes

Class A	1 – 127	(Network 127 is reserved for loopback and internal testing)	
	Leading bit pattern	0	00000000.00000000.00000000.00000000 Network . Host . Host . Host
Class B	128 – 191	Leading bit pattern	10
			10000000.00000000.00000000.00000000 Network . Network . Host . Host
Class C	192 – 223	Leading bit pattern	110
			11000000.00000000.00000000.00000000 Network . Network . Network . Host
Class D	224 – 239	(Reserved for multicast)	
Class E	240 – 255	(Reserved for experimental, used for research)	

Private Address Space

Class A	10.0.0.0 to 10.255.255.255
Class B	172.16.0.0 to 172.31.255.255
Class C	192.168.0.0 to 192.168.255.255

Default Subnet Masks

Class A	255.0.0.0
Class B	255.255.0.0
Class C	255.255.255.0

This workbook assumes you already have a background in subnetting. If you don't you may want to consider completing the [IP Addressing and Subnetting Workbook](#).

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Special Thanks to Melvin Baker and Jim Dorsch
for taking the time to check this workbook for errors,
and to everyone who has sent in suggestions to improve the series.

Workbooks included in the series:

IP Addressing and Subnetting Workbooks
ACLs - Access Lists Workbooks
VLSM Variable-Length Subnet Mask Workbooks

Instructors (and anyone else for that matter) please do not post the Instructors version on public websites. When you do this you are giving everyone else worldwide the answers. Yes, students look for answers this way. It also discourages others; myself included, from posting high quality materials.

Inside Cover

Classful vs. Classless Subnetting

When you're subnetting an IP address for a network you have two options: classful and classless. Classful subnetting is the simplest method. It tends to be the most wasteful because it uses more addresses than are necessary. In classful subnetting you use the same subnet mask for each subnet, and all the subnets have the same number of addresses in them.

Classless addressing allows you to use different subnet masks and create subnets tailored to the number of users in each group. This technique is referred to as VLSM, Variable Length Subnet Masks.

What is VLSM

Variable Length Subnet Masks allow you a much tighter control over your addressing scheme. If you use a class C address with a default subnet mask you end up with one subnet containing 256 addresses. By using VLSM you can adjust the number of subnets and number of addresses depending on the specific needs of your network. The same rules apply to a class A or B addresses.

VLSM is supported by the following protocols: RIP version 2, OSPF, EIGRP, Dual IS-IS, and BGP. You need to configure your router for Variable Length Subnet Masks by setting up one of these protocols. Then configure the subnet masks of the various interfaces in the IP address interface sub-command.

Benefits of VLSM

- Allows efficient use of address space
- Allows the use of multiple subnet mask lengths
- Breaks up an address block into smaller custom blocks
- Allows for route summarization
- Provides more flexibility in network design
- Supports hierarchical enterprise networks

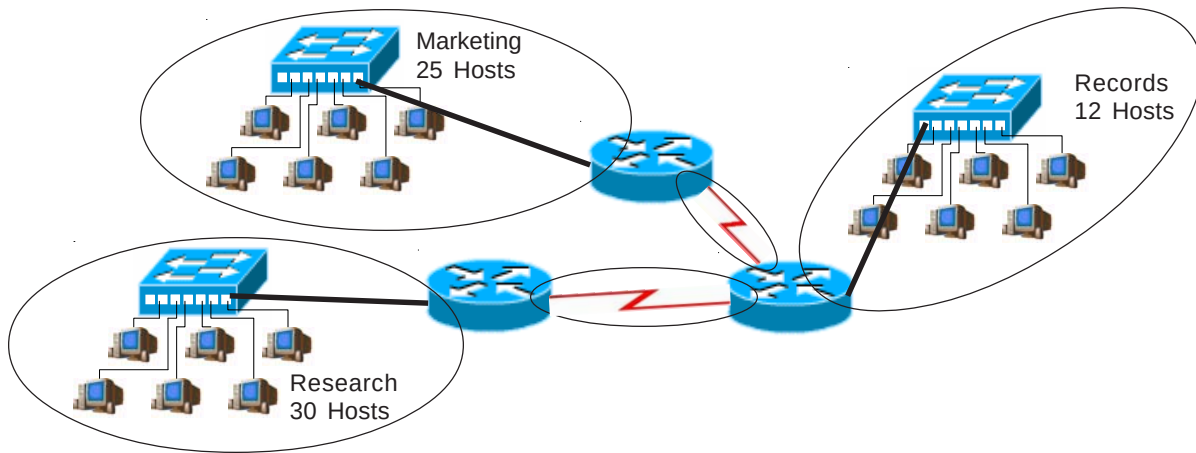
This workbook explores three different methods to figure out sub-subnets: the box method, the circle method, and a VLSM chart.

Classful Subnetting Example

When you're subnetting an IP address for a network you have two options: classful and classless. Classful subnetting is the simplest method. It also tends to be the most wasteful because it uses more addresses than are necessary. In classful subnetting you use the same subnet mask for each subnet, and all the subnets have the same number of addresses in them.

In this example you need five subnets, each one containing 30 hosts. The serial connections only require two address each so you are wasting 28 usable addresses in each of the serial subnet ranges.

IP Address: 192.168.1.0



The Box Method for visualizing subnets

Classful Subnet Ranges

192.168.1.0	to	192.168.1.31	/27
192.168.1.32	to	192.168.1.63	/27
192.168.1.64	to	192.168.1.95	/27
192.168.1.96	to	192.168.1.127	/27
192.168.1.128	to	192.168.1.159	/27
192.168.1.160	to	192.168.1.191	/27
192.168.1.192	to	192.168.1.223	/27
192.168.1.224	to	192.168.1.255	/27
			/27
255.255.255.224			
32 Hosts			
8 Subnets			

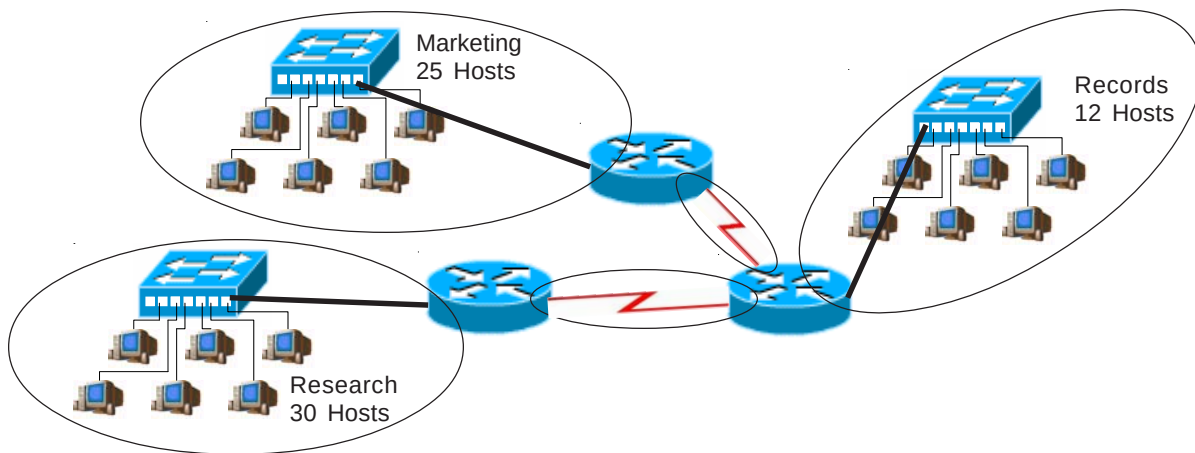
0 31	32 63	128 159	160 191
64 95	96 127	192 223	224 255

Classless Subnetting Example

Classless addressing allows you to use different subnet masks and create subnets tailored to the number of users in each subnetwork. There are fewer wasted IP addresses using smaller subnets.

In this example you need at total of five subnets, two containing 30 hosts, one containing 12 hosts, and two serial connections that only require two usable addresses each.

IP Address: 192.168.1.0

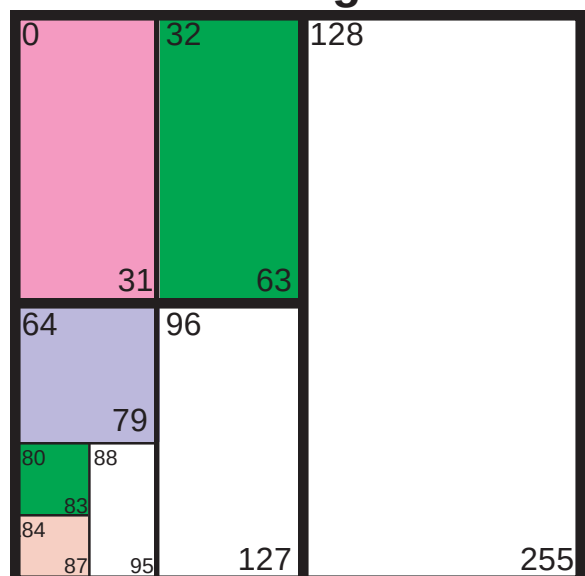


By adjusting the subnet masks you can cut your address usage by almost half in this example. This type of subnetting requires a network protocol which will support it such as: RIP version 2, EIGRP, OSPF, or BGP.

The Box Method for visualizing subnets

Classless Subnet Ranges

192.168.1.0	to	192.168.1.31	/27
192.168.1.32	to	192.168.1.63	/27
192.168.1.64	to	192.168.1.79	/28
192.168.1.80	to	192.168.1.87	/30
192.168.1.88	to	192.168.1.95	/29
192.168.1.96	to	192.168.1.127	/27
192.168.1.128	to	192.168.1.255	/25

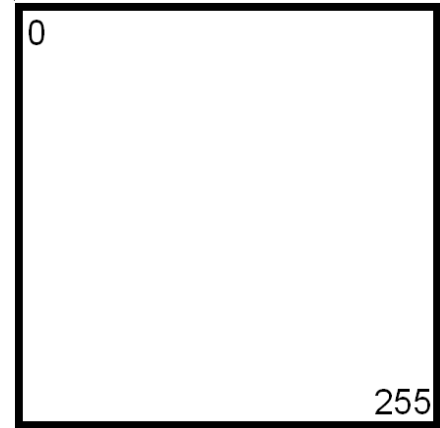


Visualizing Subnets Using The Box Method

The box method is a simple way to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

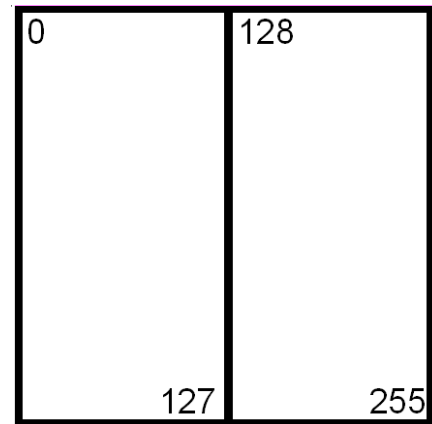
Start with a square. The whole square is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



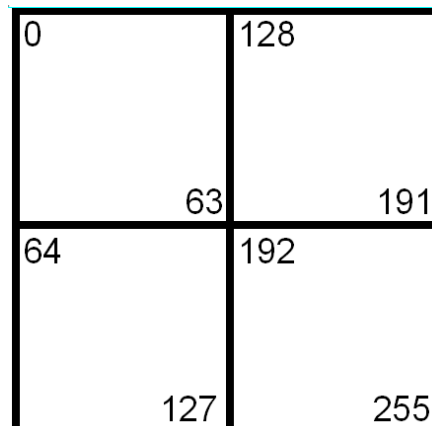
Split the box in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



Divide the box into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



Split each individual square and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets

0	32	128	160
31	63	159	191
64	96	192	224
95	127	223	255

Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets

0	32	128	160
15	47	143	175
16	48	144	176
31	63	159	191
64	96	192	224
79	111	207	239
80	112	208	240
95	127	223	255

The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets

0	8	32	40	128	136	160	168
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
87	95	119	127	215	223	247	255

The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets

0	8	32	40	128	136	160	168
3	11	35	43	131	139	163	171
4	12	36	44	132	140	164	172
7	15	39	47	135	143	167	175
16	24	48	56	144	152	176	184
19	27	51	59	147	155	179	187
20	28	52	60	148	156	180	188
23	31	55	63	151	159	183	191
64	72	96	104	192	200	224	232
67	75	99	107	195	203	227	235
68	76	100	108	196	204	228	236
71	79	103	111	199	207	231	239
80	88	112	120	208	216	240	248
83	91	115	123	211	219	243	251
84	92	116	124	212	220	244	252
87	95	119	127	215	223	247	255

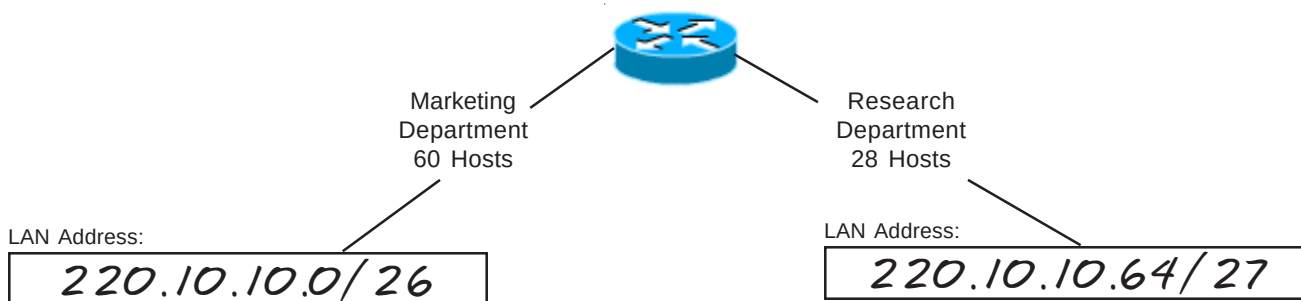
VLSM Addressing

Box Method

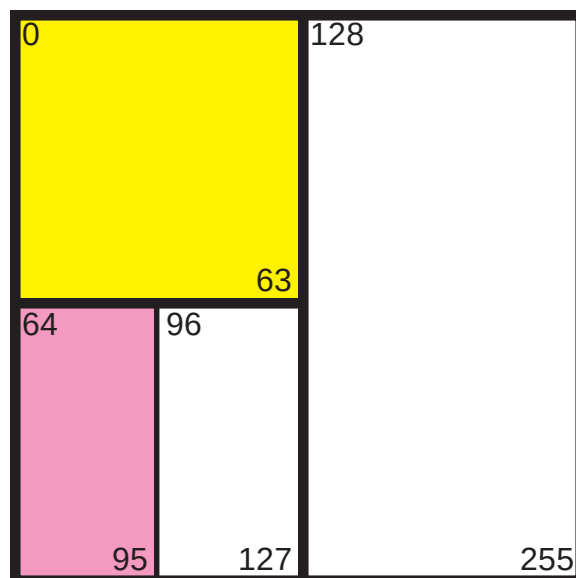
(Sample)

Problem 1

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.



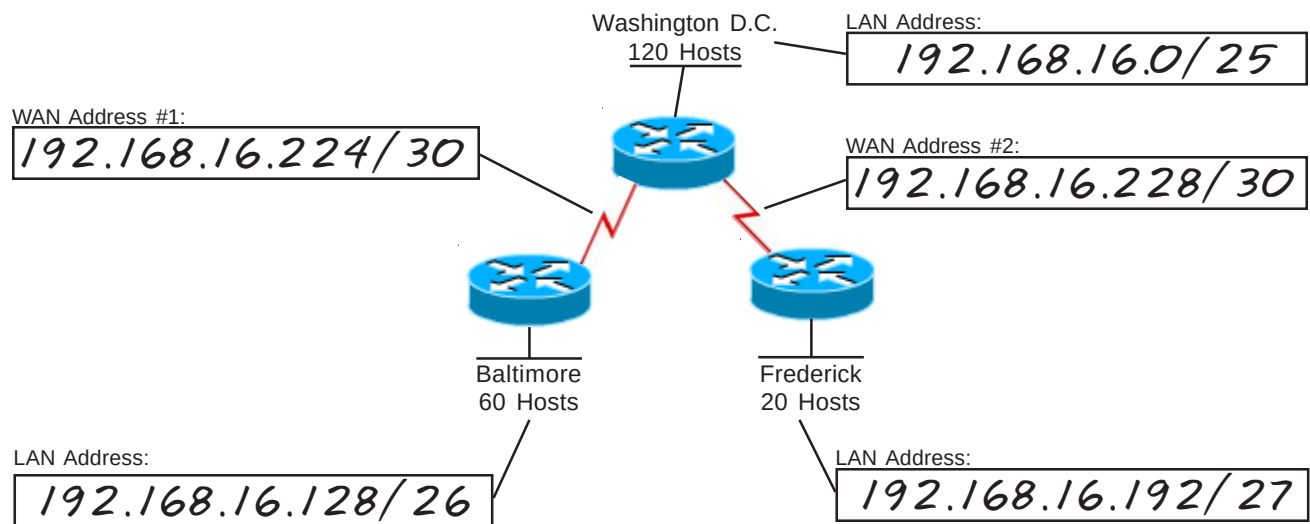
VLSM Addressing

Box Method

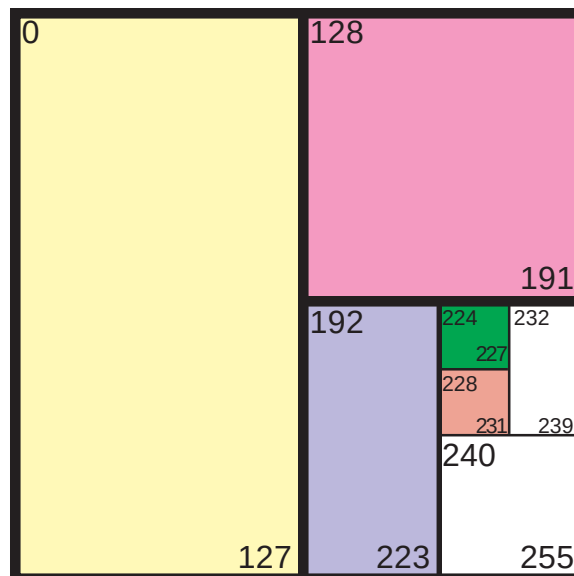
(Sample)

Problem 2

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

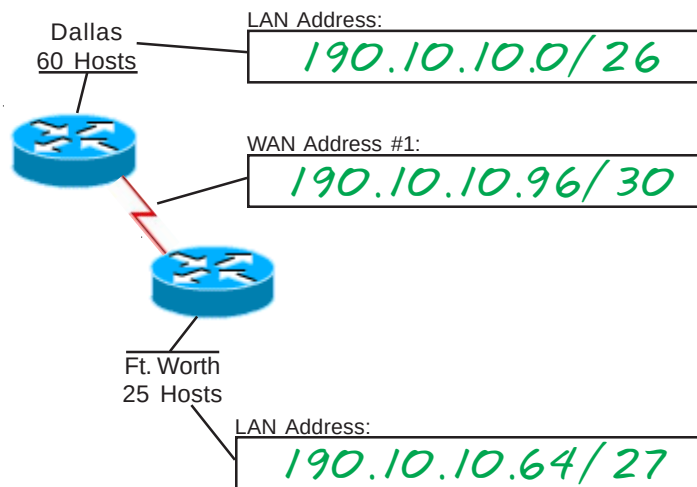


VLSM Addressing

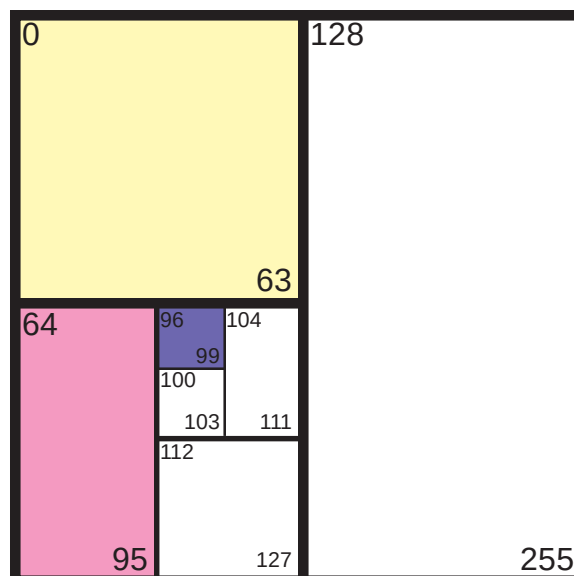
Box Method

Problem 3

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 190.10.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

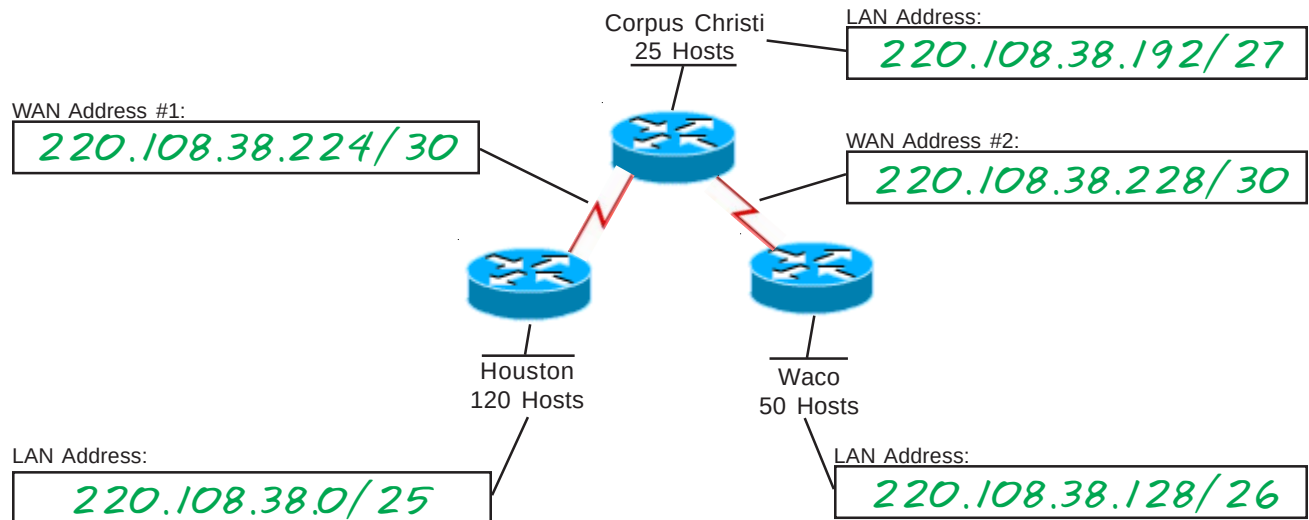


VLSM Addressing

Box Method

Problem 4

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 220.108.38.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

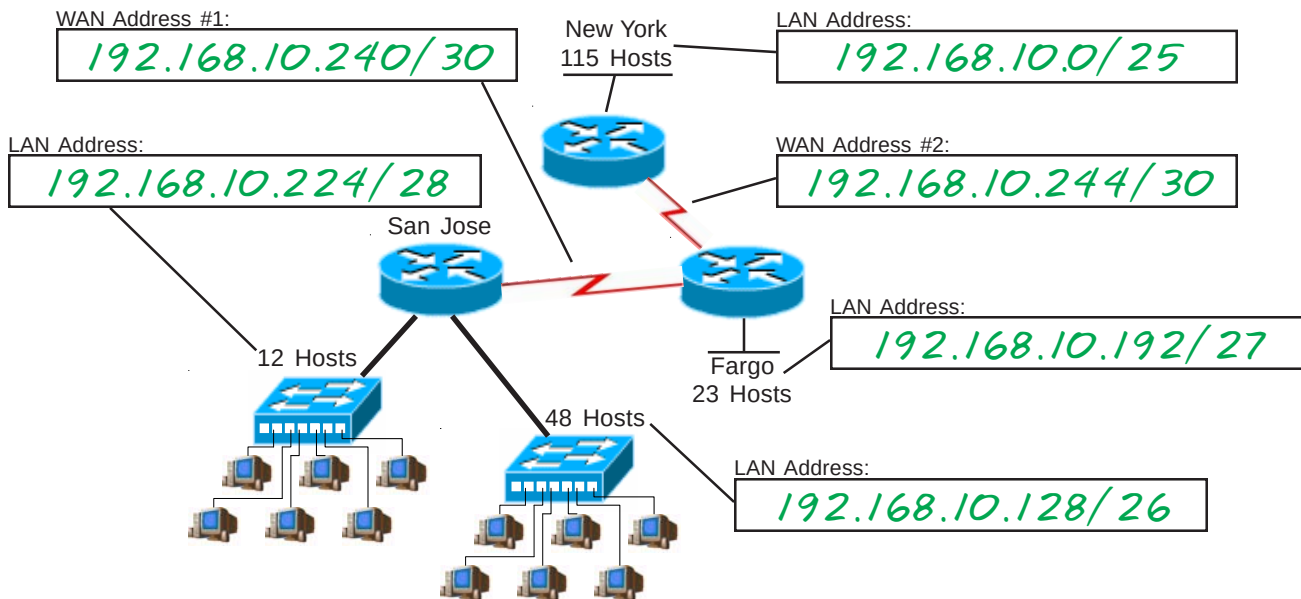
0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247

VLSM Addressing

Box Method

Problem 5

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.10.0. Remember to start with your largest groups first.



Color in the squares used with different shades to highlight each subnet.

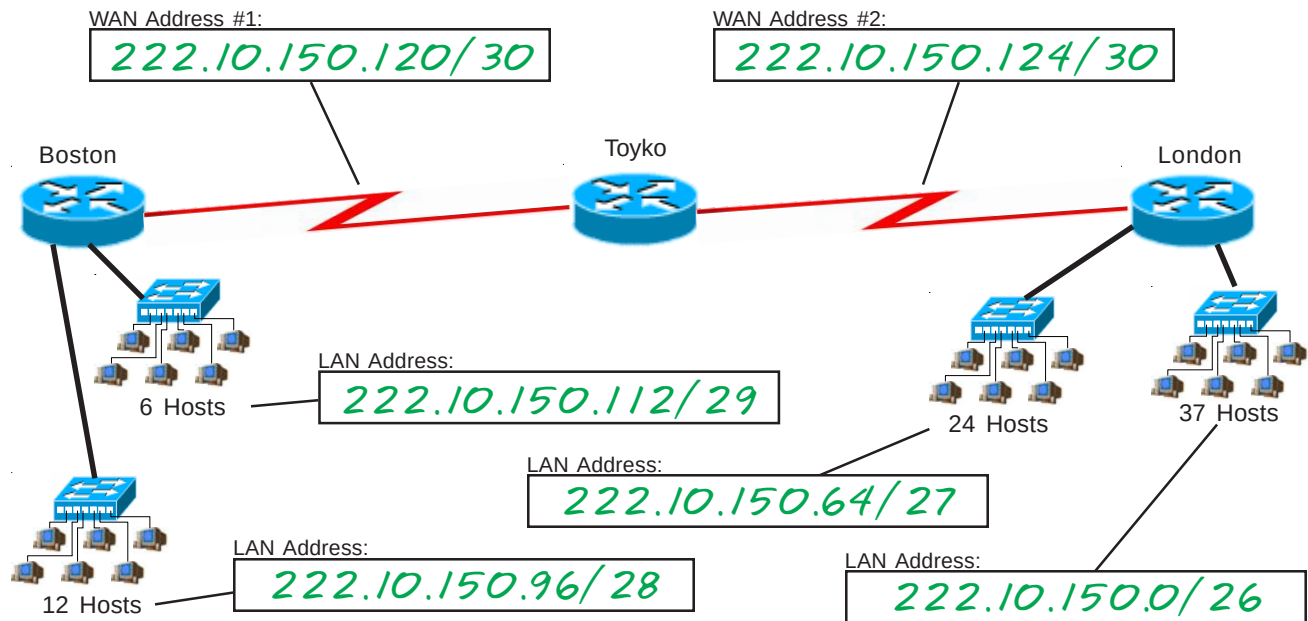
0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247

VLSM Addressing

Box Method

Problem 6

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 222.10.150.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

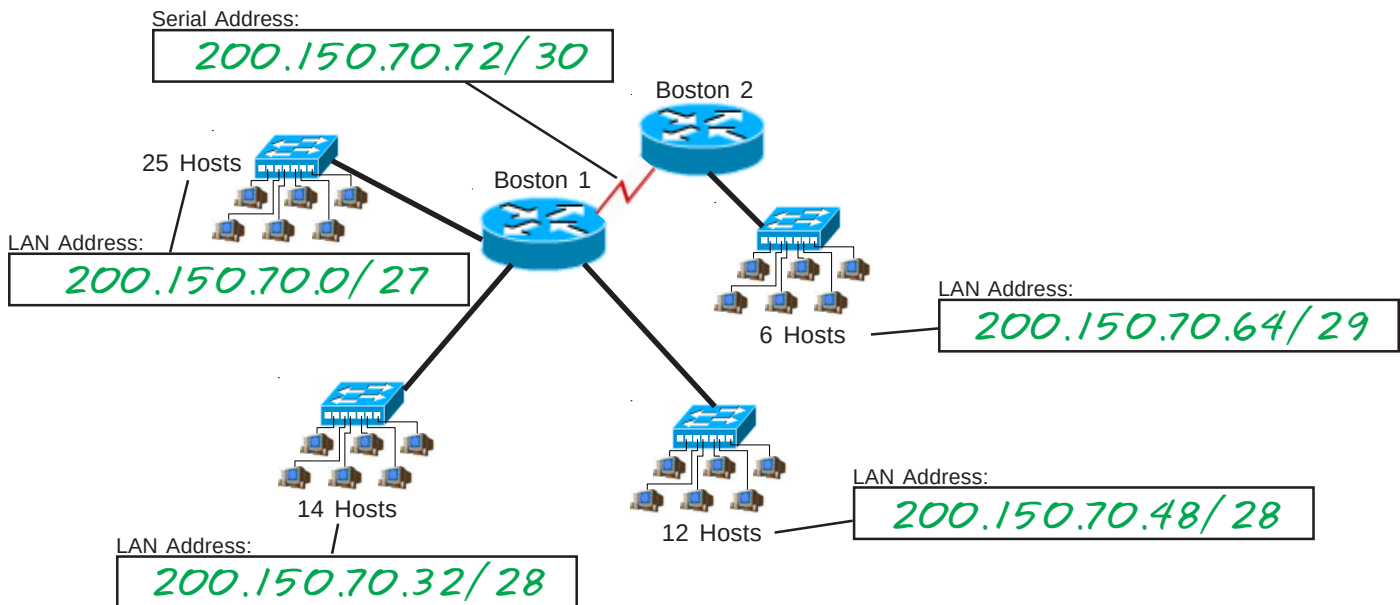
0	32	128	160
31	63	159	191
64	96	192	224
95	111	223	255
	112		
	120		
	123		
	124		
	119		
	127		

VLSM Addressing

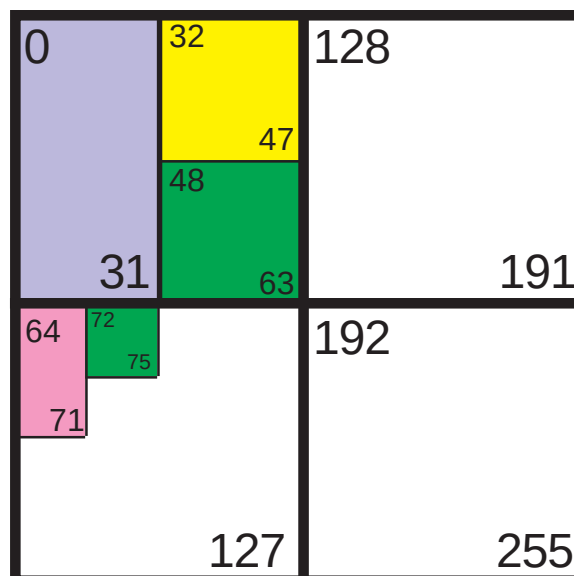
Box Method

Problem 7

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and subnet mask in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 200.150.70.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

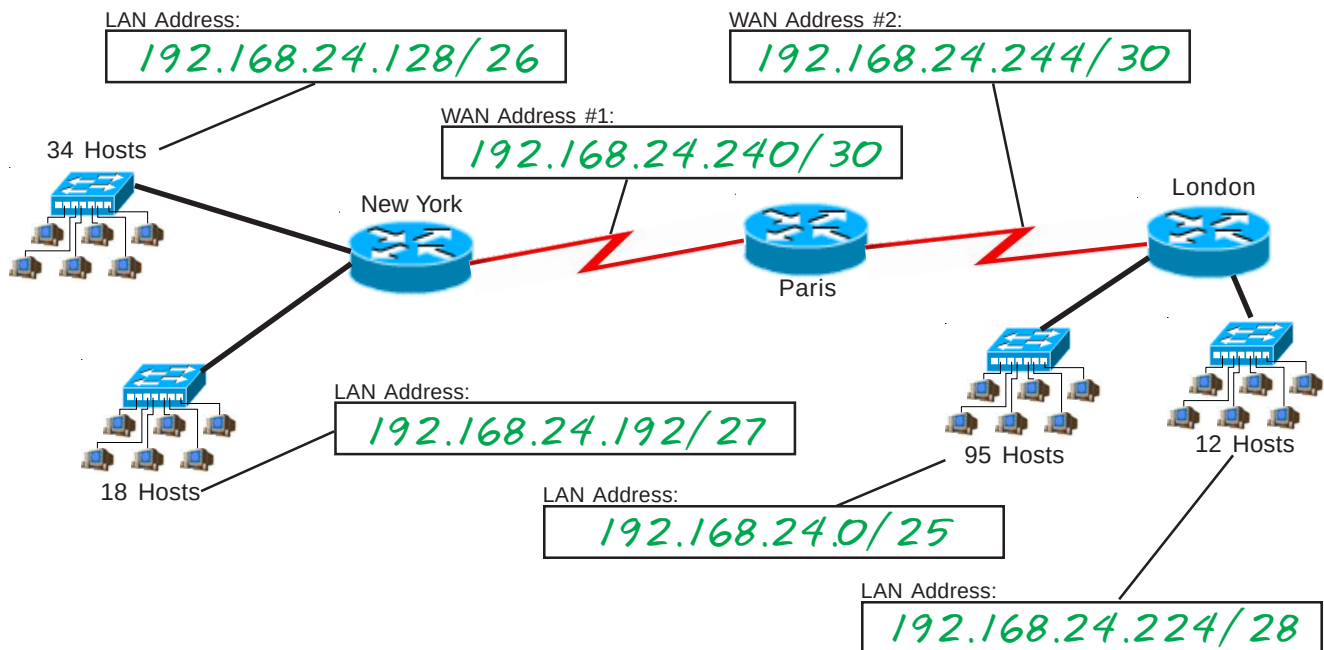


VLSM Addressing

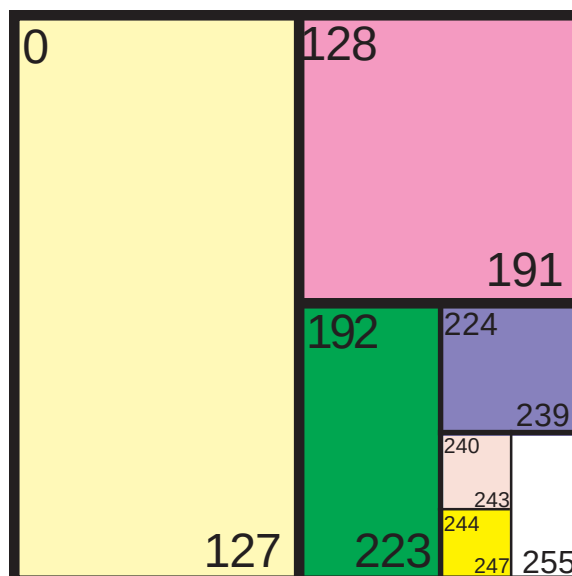
Box Method

Problem 8

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and subnet mask in the boxes below, color or shade the sub-subnets used in the box. This company will be using the class C address 192.168.24.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used squares with different shades to highlight each subnet.

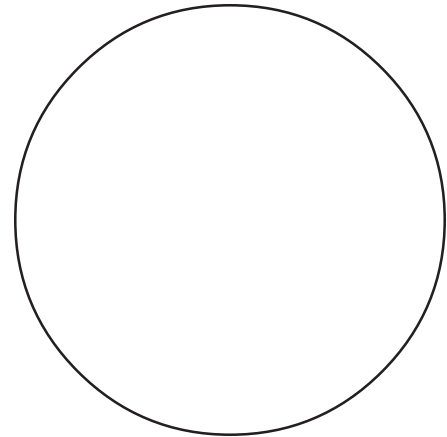


Visualizing Subnets Using The Circle Method

The circle method is another method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the different sections of the circle you can easily break up your subnets without overlapping your addresses. You adjust each subnet to the correct size needed.

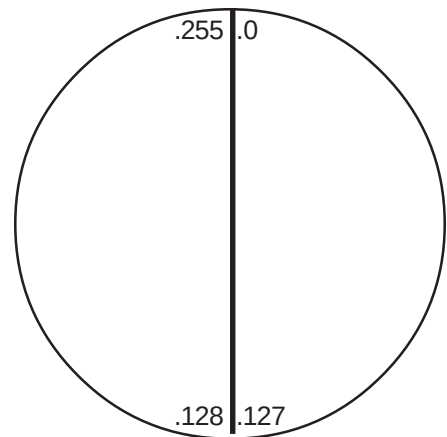
Start with a circle. The whole circle is a single subnet comprised of 256 addresses.

/24
255.255.255.0
256 Hosts
1 Subnet



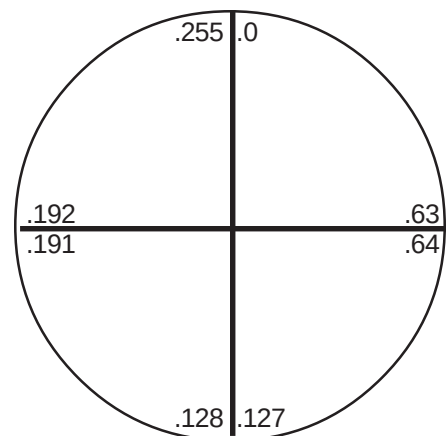
Split the circle in half and you get two subnets with 128 addresses.

/25
255.255.255.128
128 Hosts
2 Subnets



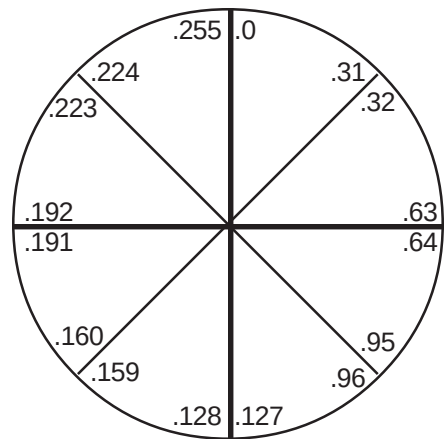
Divide the circle into quarters and you get four subnets with 64 addresses.

/26
255.255.255.192
64 Hosts
4 Subnets



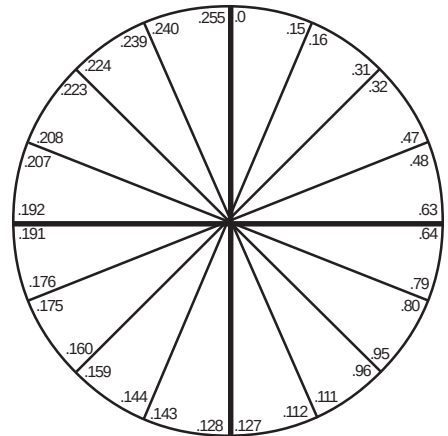
Split each quarter and you get eight subnets with 32 addresses.

/27
255.255.255.224
32 Hosts
8 Subnets



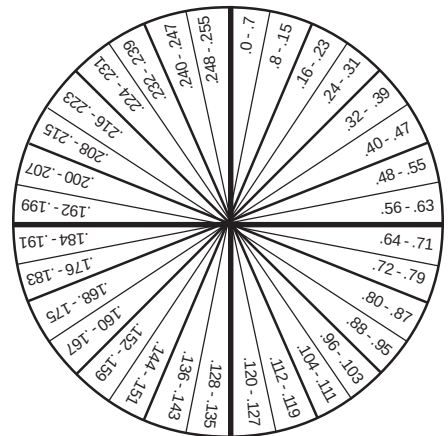
Split the boxes in half again and you get sixteen subnets with sixteen addresses.

/28
255.255.255.240
16 Hosts
16 Subnets



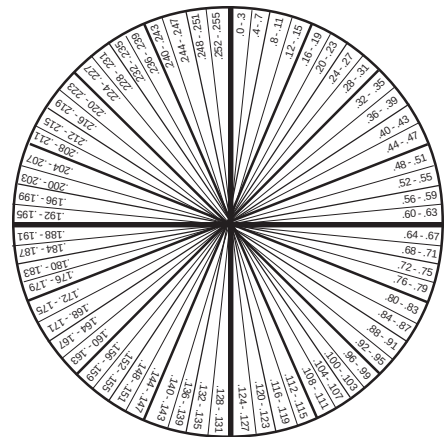
The next split gives you thirty two subnets with eight addresses.

/29
255.255.255.248
8 Hosts
32 Subnets



The last split gives sixty four subnets with four addresses each.

/30
255.255.255.252
4 Hosts
64 Subnets



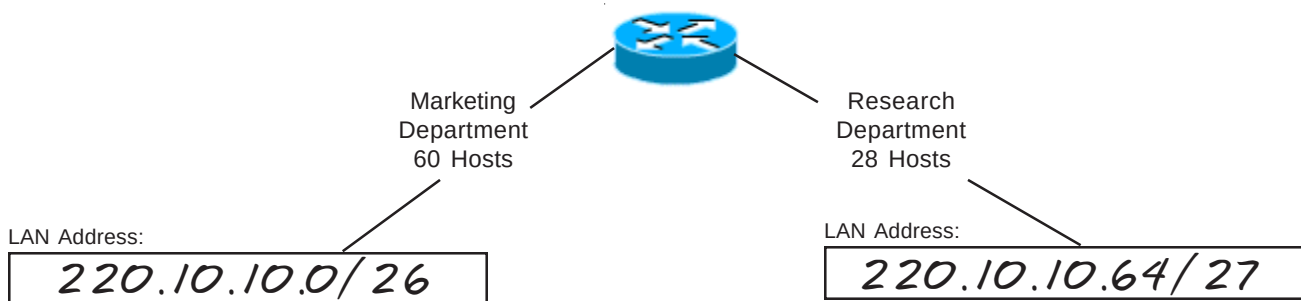
VLSM Addressing

Circle Method

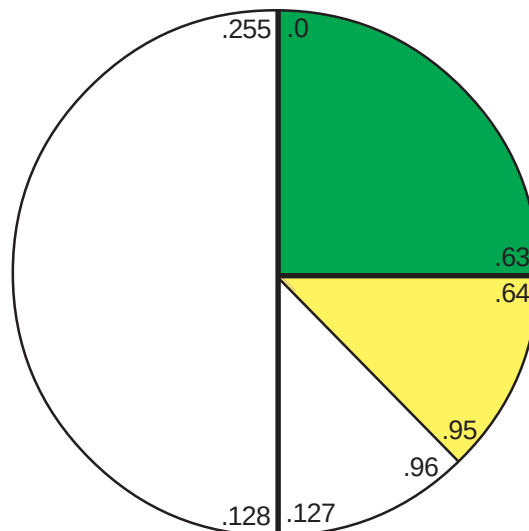
(Sample)

Problem 9

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the circle below, color or shade the sub-subnets used. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Color in the necessary circle sections used with different shades to highlight each subnet.



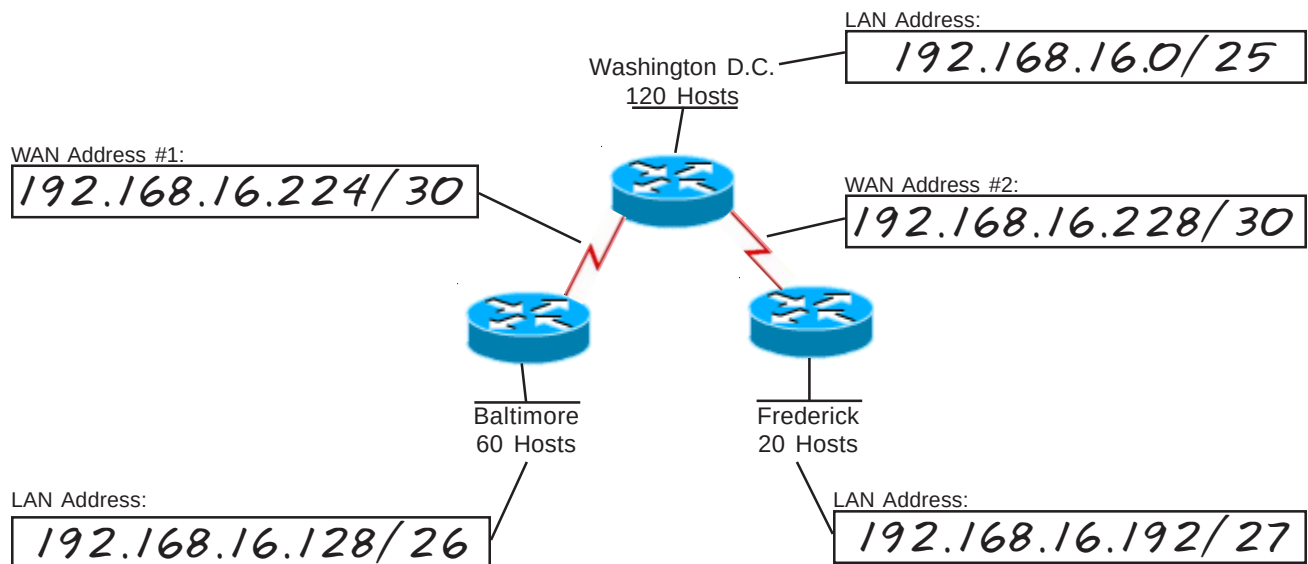
VLSM Addressing

Circle Method

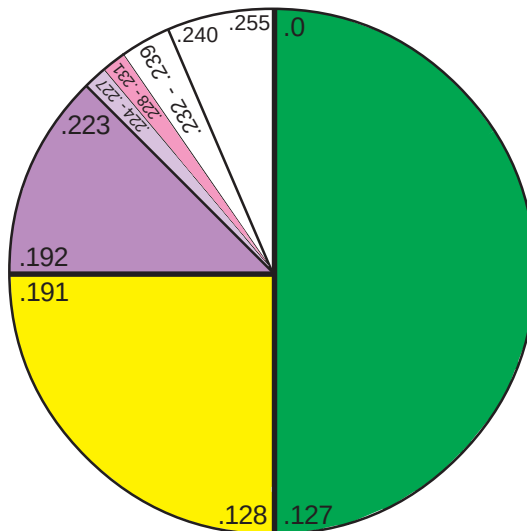
(Sample)

Problem 10

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the circle below, color or shade the sub-subnets used. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Color in the necessary circle sections used with different shades to highlight each subnet.



VLSM Addressing

Circle Method

Problem 11

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the circle below, color or shade the sub-subnets used. This business will be using the class C address 200.20.20.0. Remember to start with your largest groups first.

LAN Address:

200.20.20.0/25

Marketing
Department
100 Hosts

Marketing
Department
50 Hosts

Department
25 Hosts

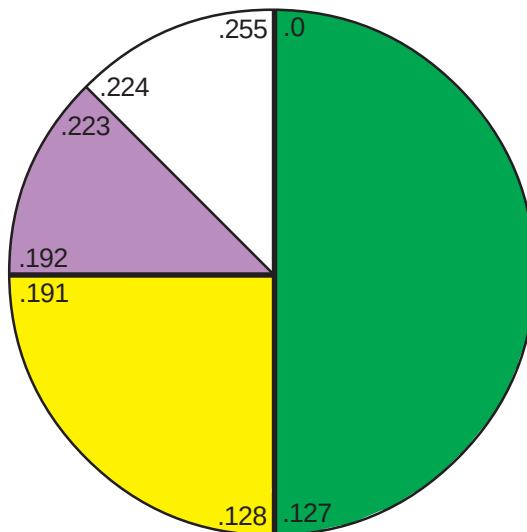
LAN Address:

200.20.20.128/26

LAN Address:

200.20.20.192/27

Color in the necessary circle sections used with different shades to highlight each subnet.

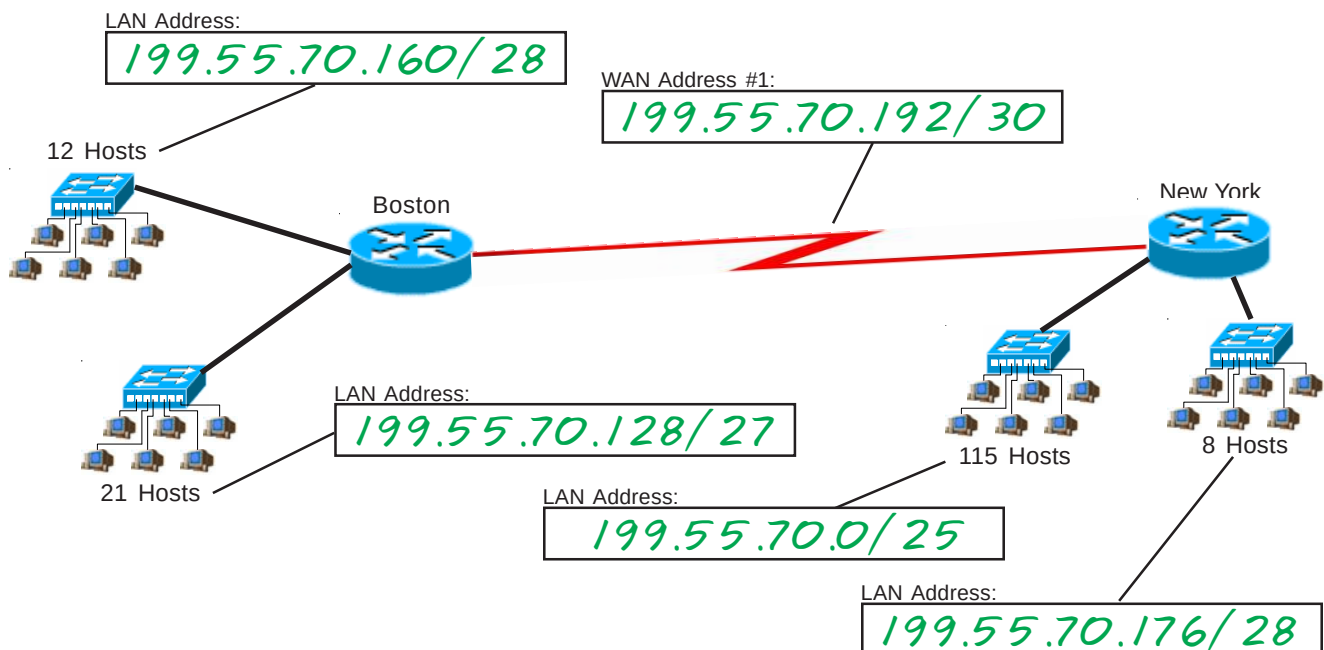


VLSM Addressing

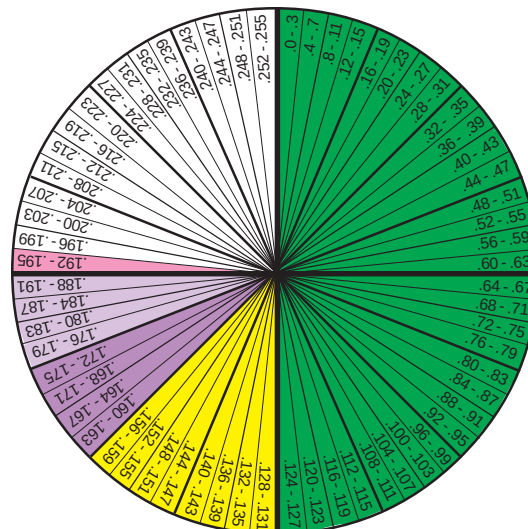
Circle Method

Problem 12

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the circle below, color or shade the sub-subnets used. This company will be using the class C address 199.55.70.0. Remember to start with your largest groups first.



Color in the necessary circle sections used with different shades to highlight each subnet.

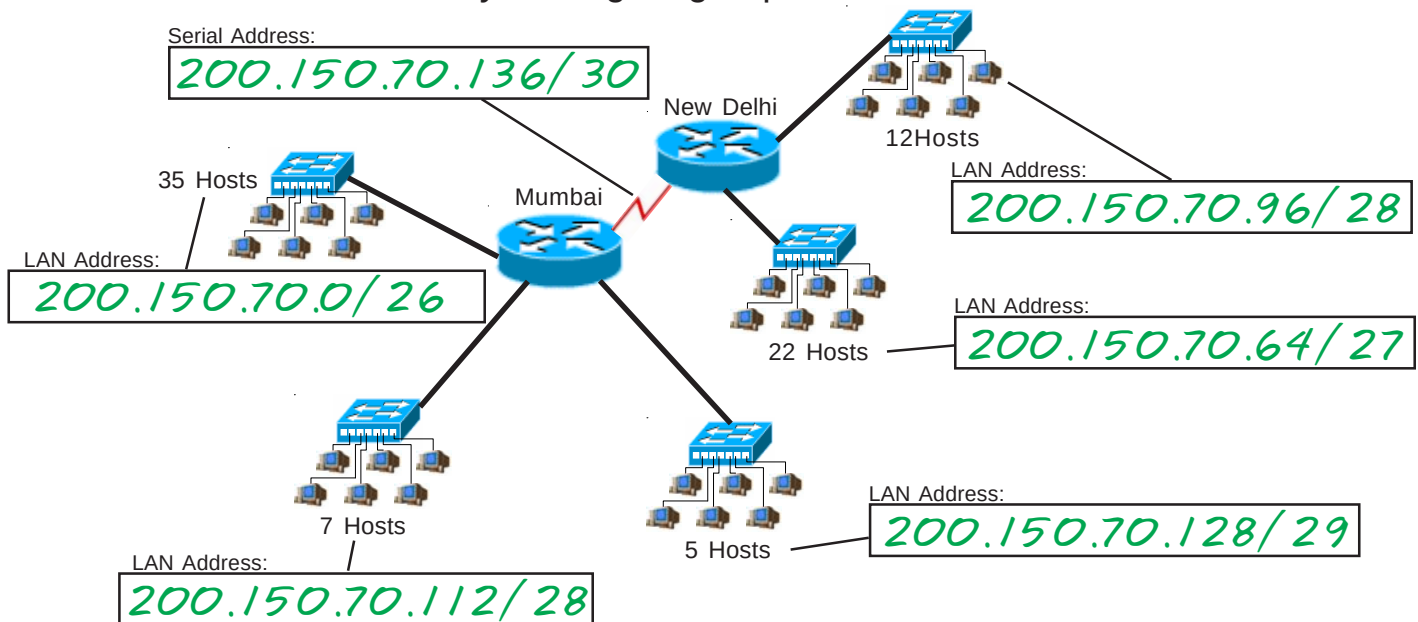


VLSM Addressing

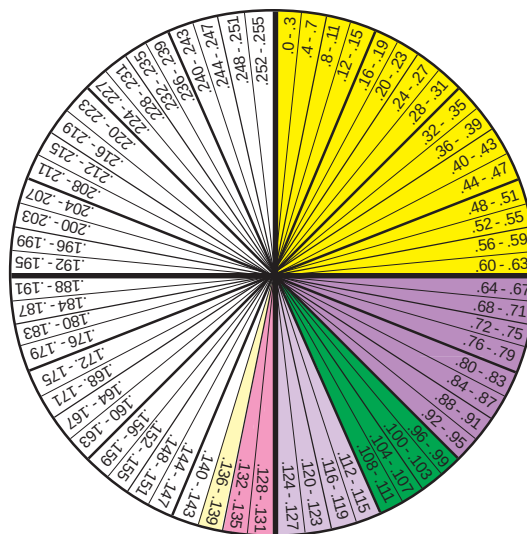
Circle Method

Problem 13

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the circle below, color or shade the sub-subnets used. This company will be using the class C address 200.150.70.0. Remember to start with your largest groups first.



Color in the necessary circle sections used with different shades to highlight each subnet.

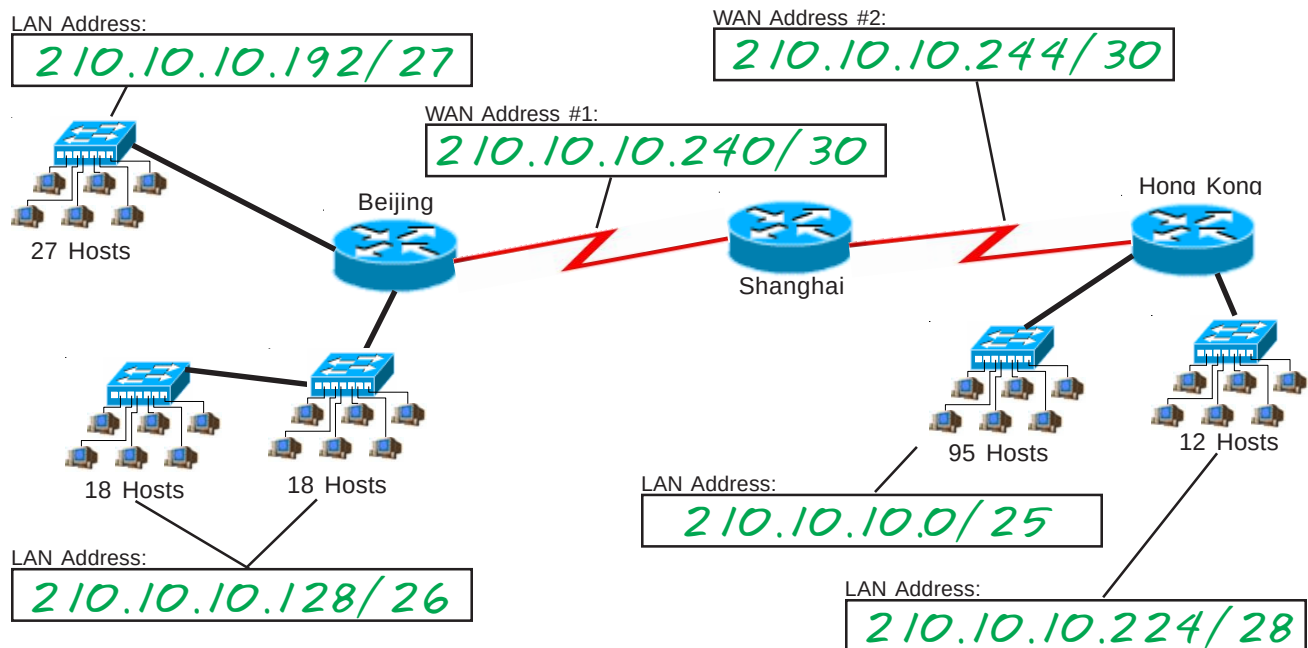


VLSM Addressing

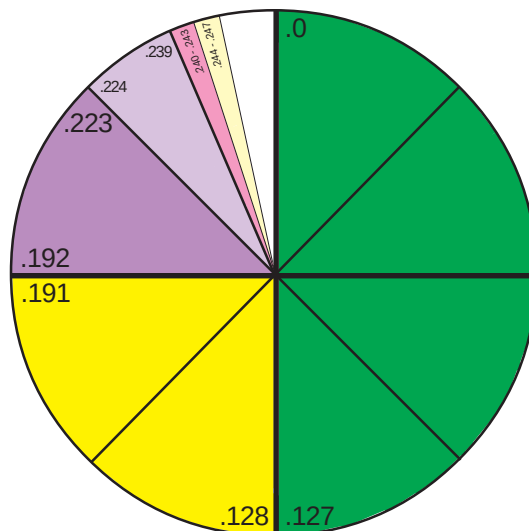
Circle Method

Problem 14

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the circle below, color or shade the sub-subnets used. This company will be using the class C address 210.10.10.0. Remember to start with your largest groups first.



Draw the necessary lines and color in the used circle sections with different shades to highlight each subnet.

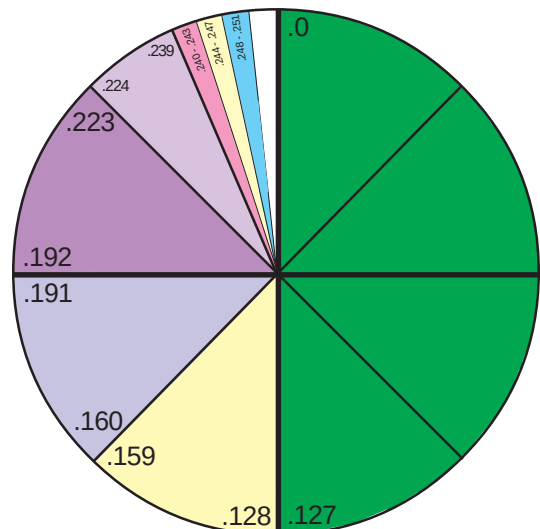
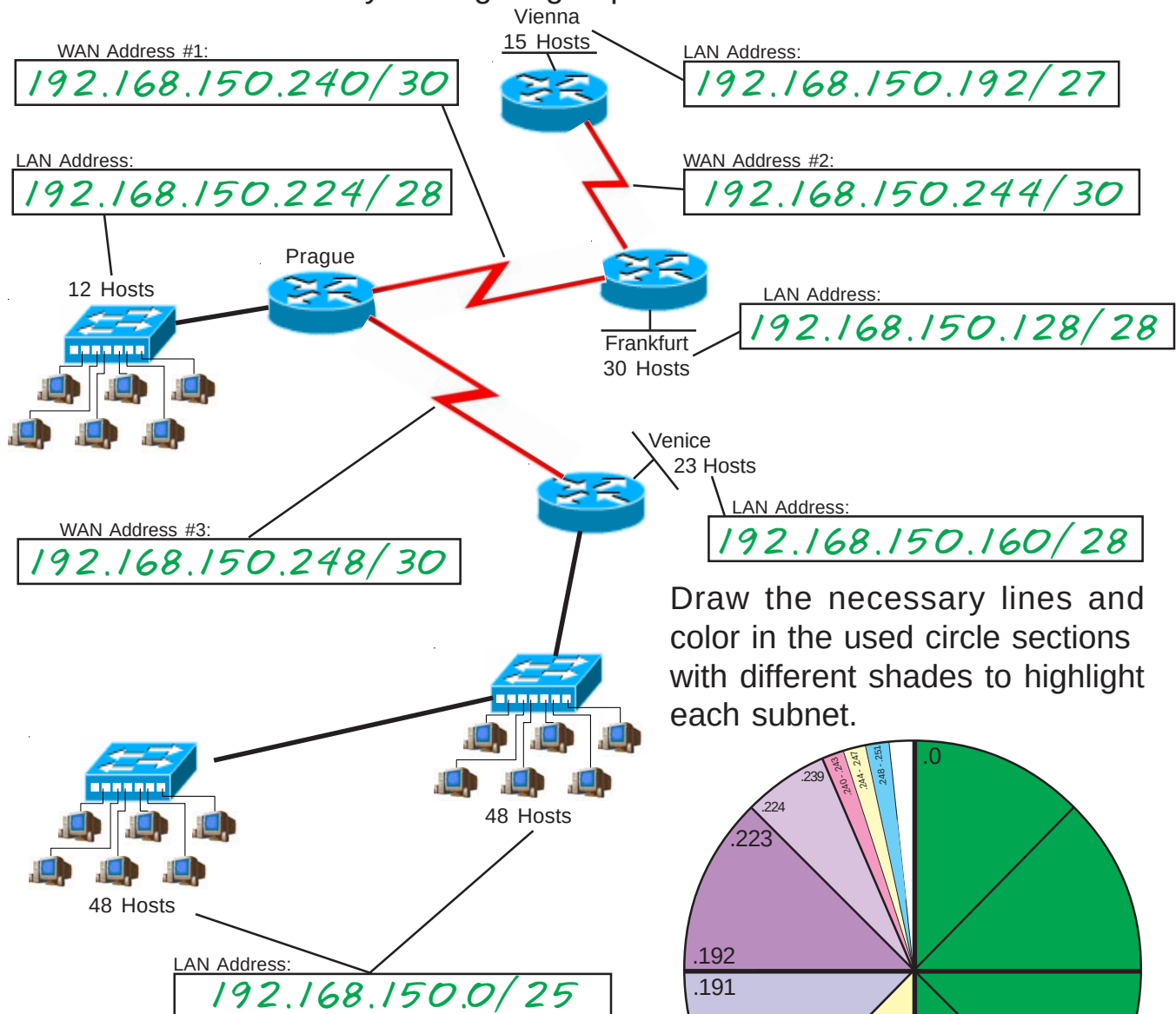


VLSM Addressing

Circle Method

Problem 15

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the circle. This company will be using the class C address 192.168.150.0. Remember to start with your largest groups first.

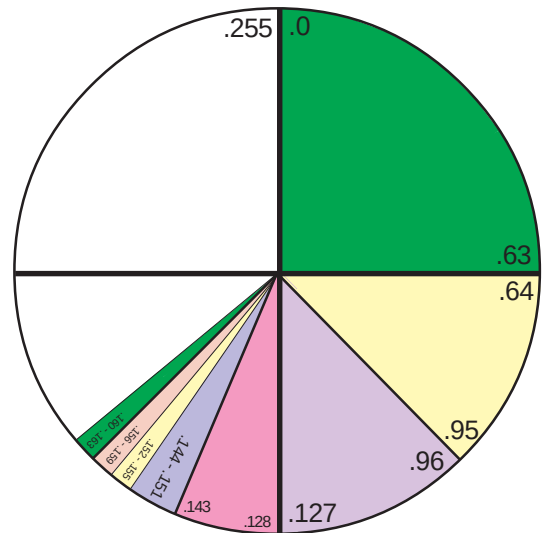
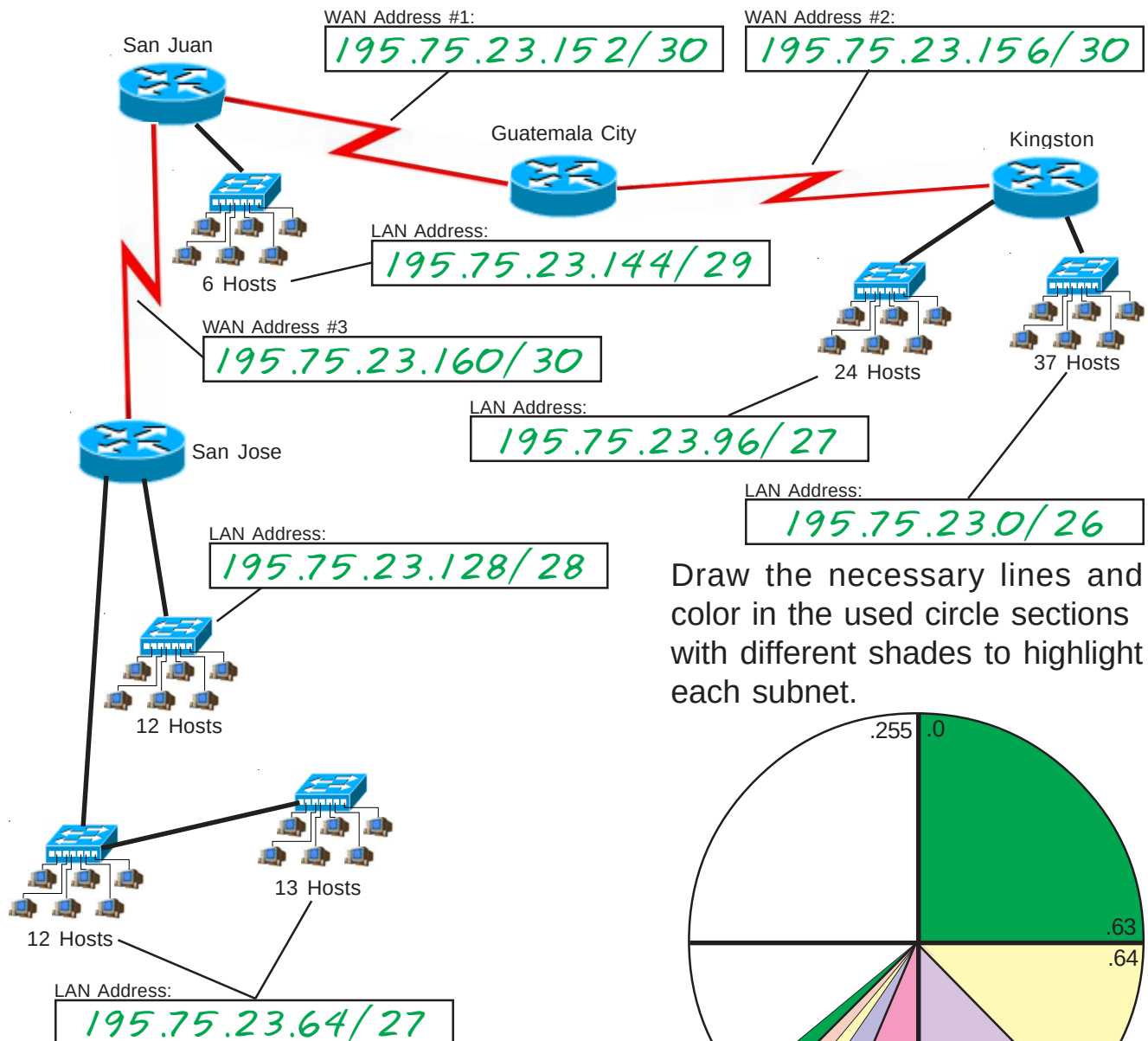


VLSM Addressing

Circle Method

Problem 16

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the circle. This company will be using the class C address 195.75.23.0. Remember to start with your largest groups first.



Visualizing Subnets Using a VLSM Chart

The VLSM chart is the third method used to visualize the breakdown of subnets and addresses into smaller sizes. By shading or coloring in the boxes you can easily break up your subnets without overlapping your addresses. You can adjust each sub-subnet to the correct size needed.

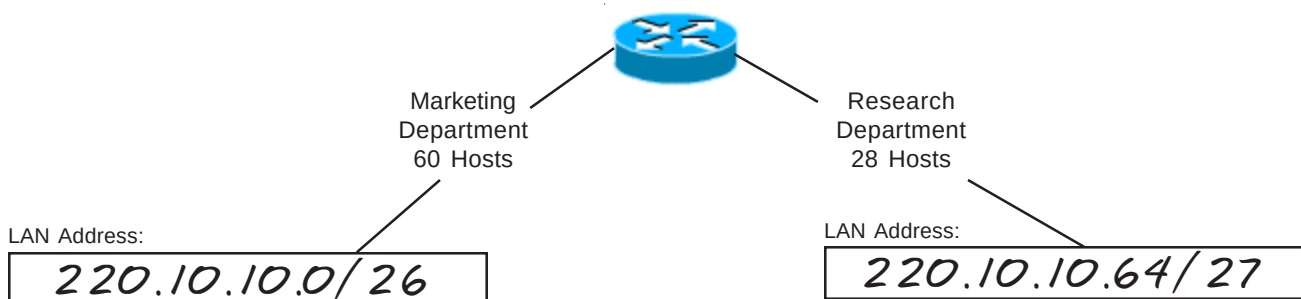
VLSM Addressing

VLSM Chart Method

(Sample)

Problem 17

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This business will be using the class C address 220.10.10.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

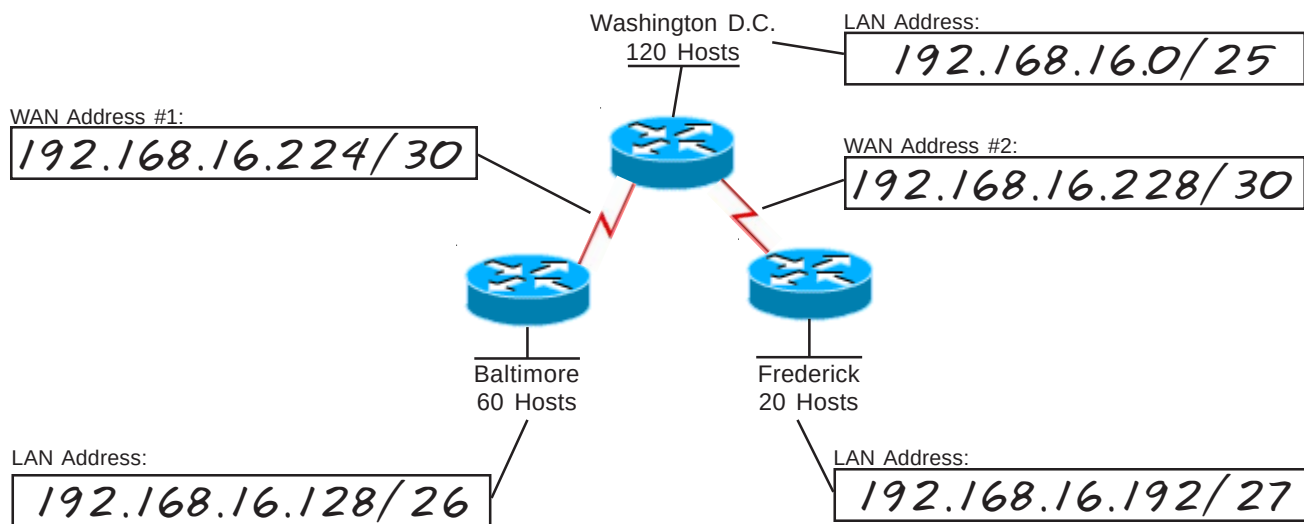
VLSM Addressing

VLSM Chart Method

(Sample)

Problem 18

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 192.168.16.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

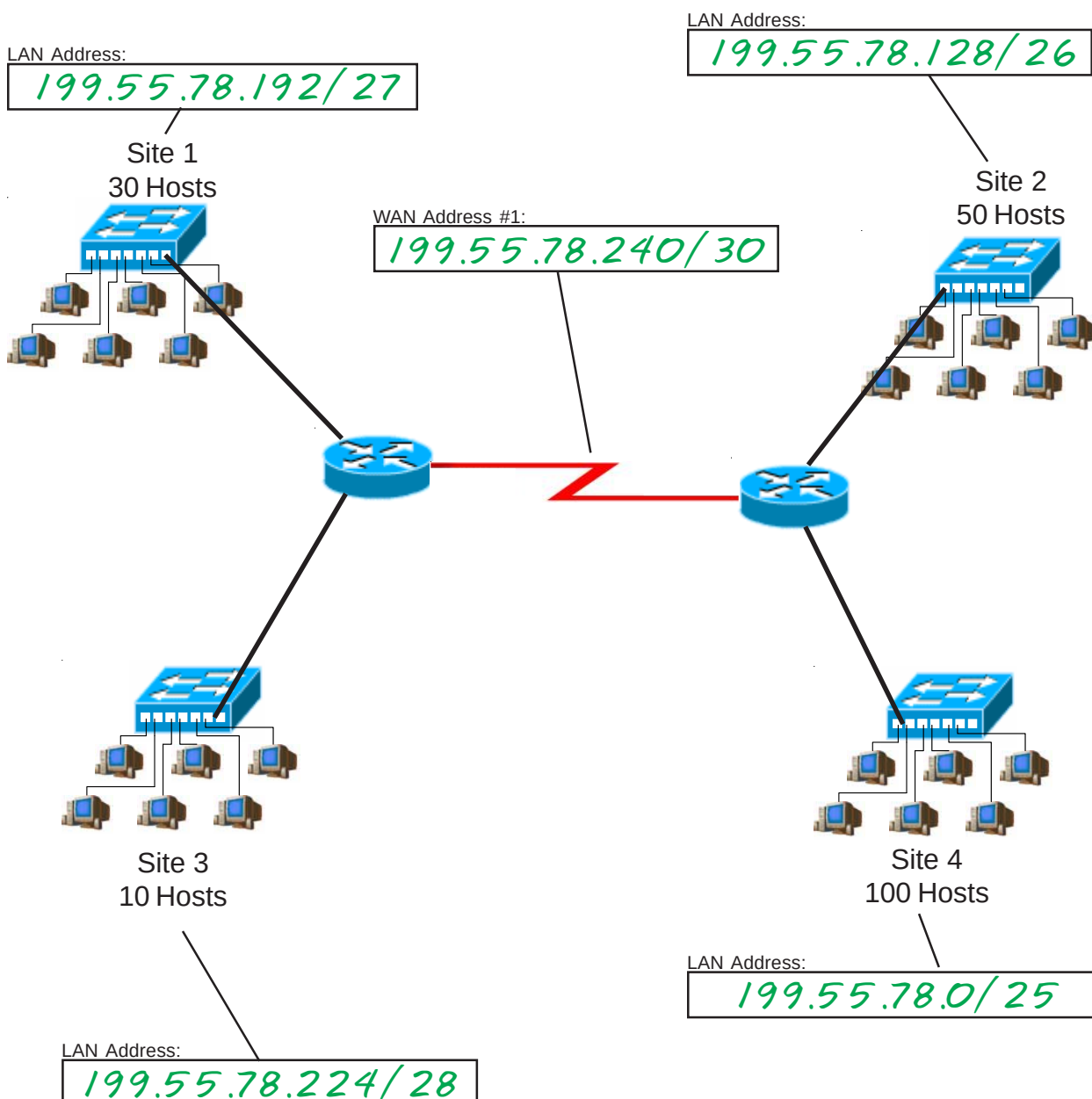
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 19

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 199.55.78.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

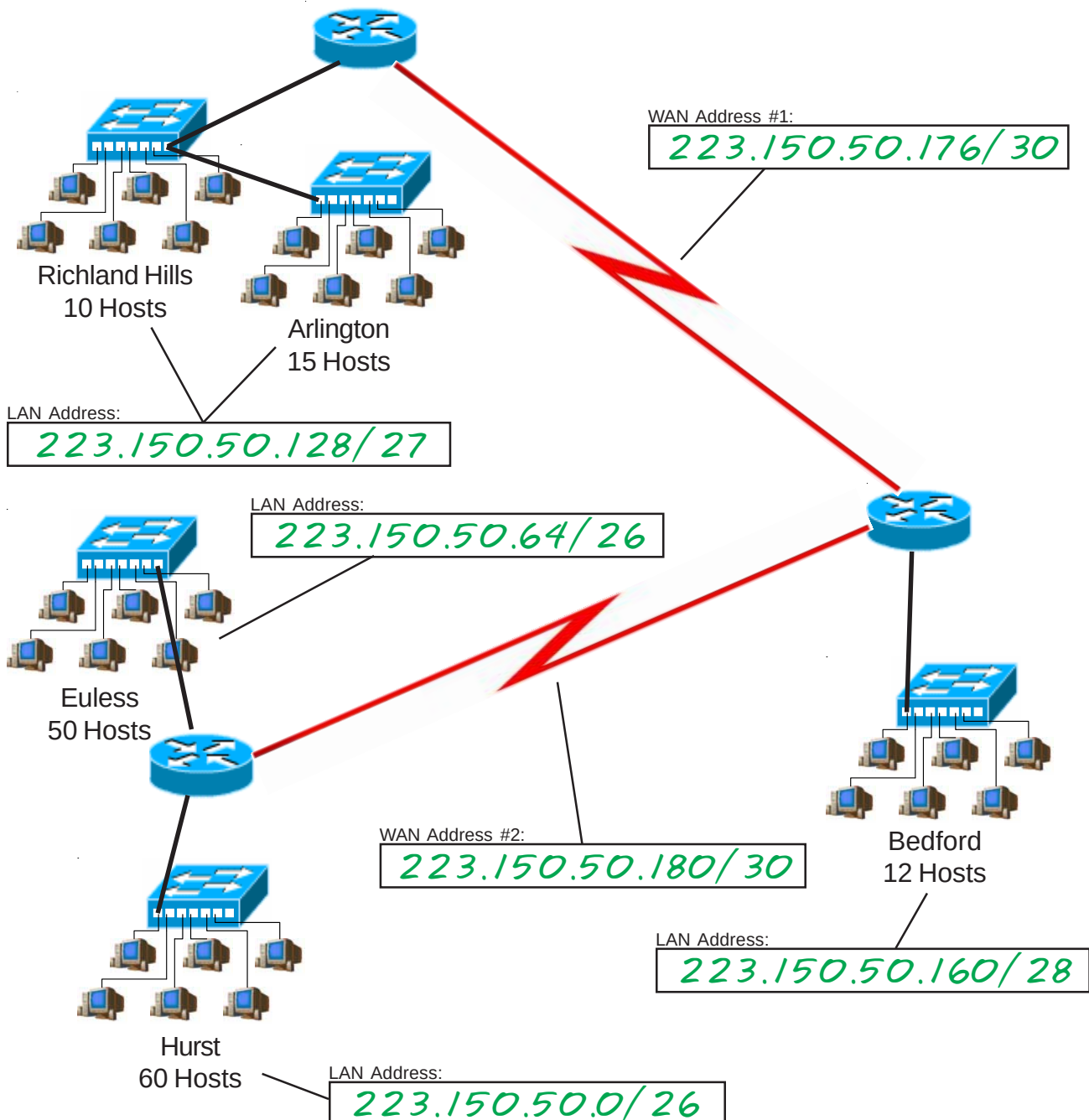
/24	/25	/26	/27	/28	/29	/30				
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts				
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7				
					16-31	8-15	8-11 12-15			
						32-47	16-23	16-19 20-23		
							24-31	24-27 28-31		
				32-63			48-63	32-39	32-35 36-39	
					40-47			40-43 44-47		
					64-79	48-55		48-51 52-55		
						56-63		56-59 60-63		
			64-127			64-95	80-95	64-71	64-67 68-71	
								96-111	72-79	72-75 76-79
					96-127				80-87	80-83 84-87
									88-95	88-91 92-95
				128-191			128-159		144-159	96-103
								104-111		104-107 108-111
					160-191			112-119		112-115 116-119
								120-127		120-123 124-127
		128-255				128-191	128-159	144-159	128-135	128-131 132-135
									136-143	136-139 140-143
					144-151				144-147 148-151	
					152-159				152-155 156-159	
				160-191	176-191		16-167	160-163 164-167		
							168-175	168-171 172-175		
							176-183	176-179 180-183		
							184-191	184-187 188-191		
			192-255	192-223	208-223	224-239	192-199	192-195 196-199		
							200-207	200-203 204-207		
							208-215	208-211 212-215		
							216-223	216-219 220-223		
				240-255	240-255	240-255	224-231	224-227 228-231		
							232-239	232-235 236-239		
							240-247	240-243 244-247		
							248-255	248-251 252-255		

VLSM Addressing

VLSM Chart Method

Problem 20

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 223.150.50.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

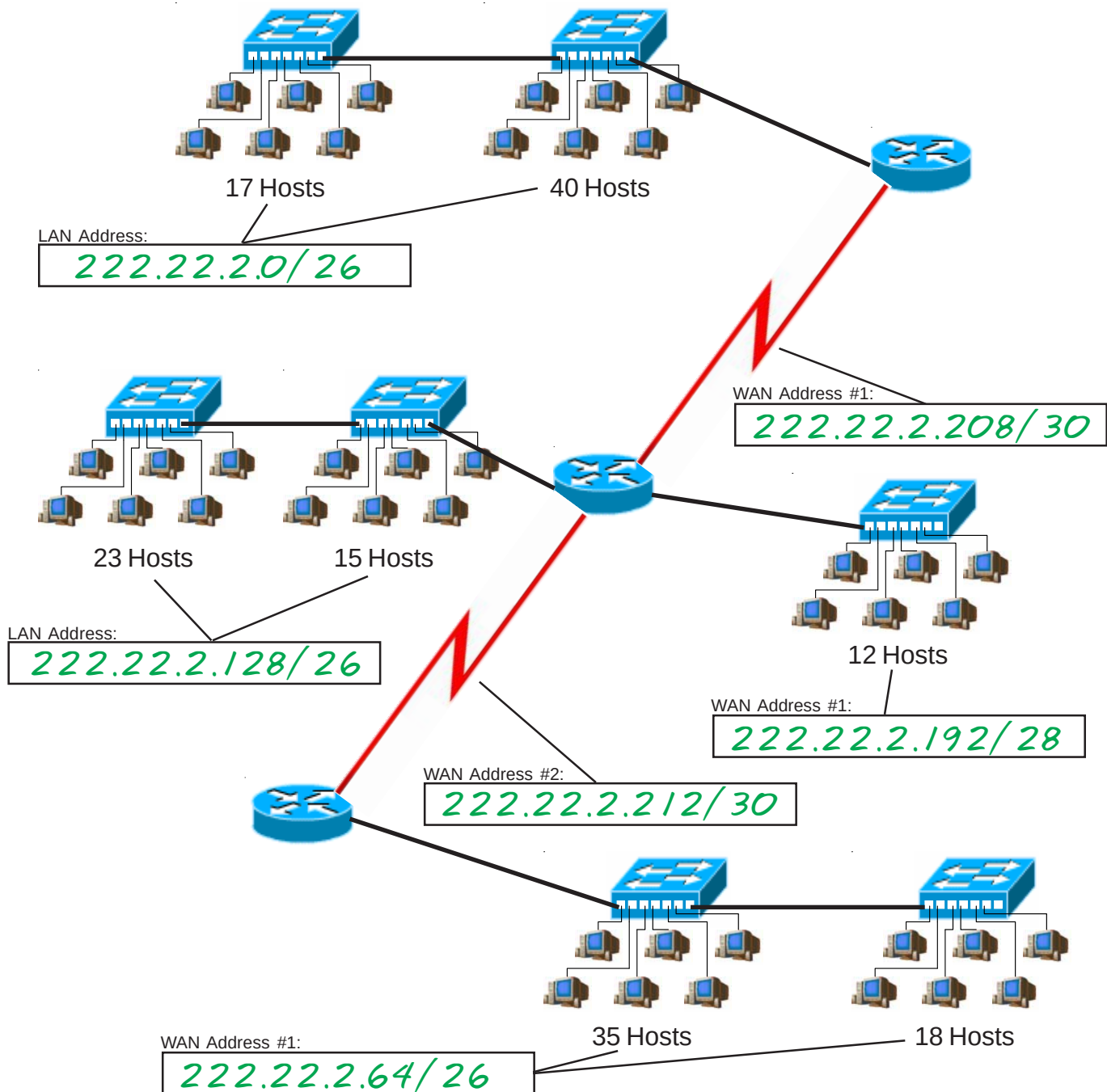
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 21

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 222.22.2.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

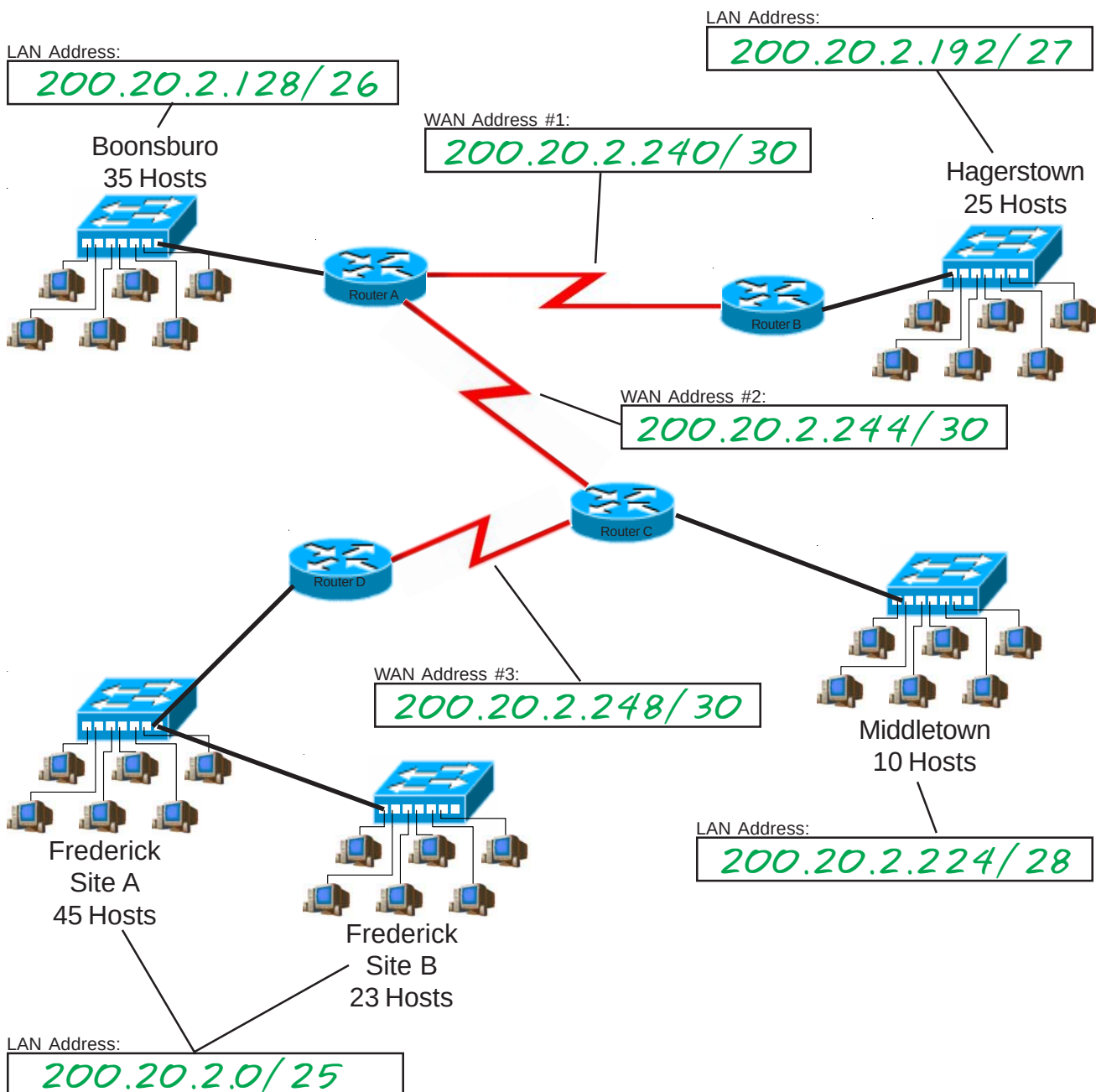
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 22

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 200.20.2.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

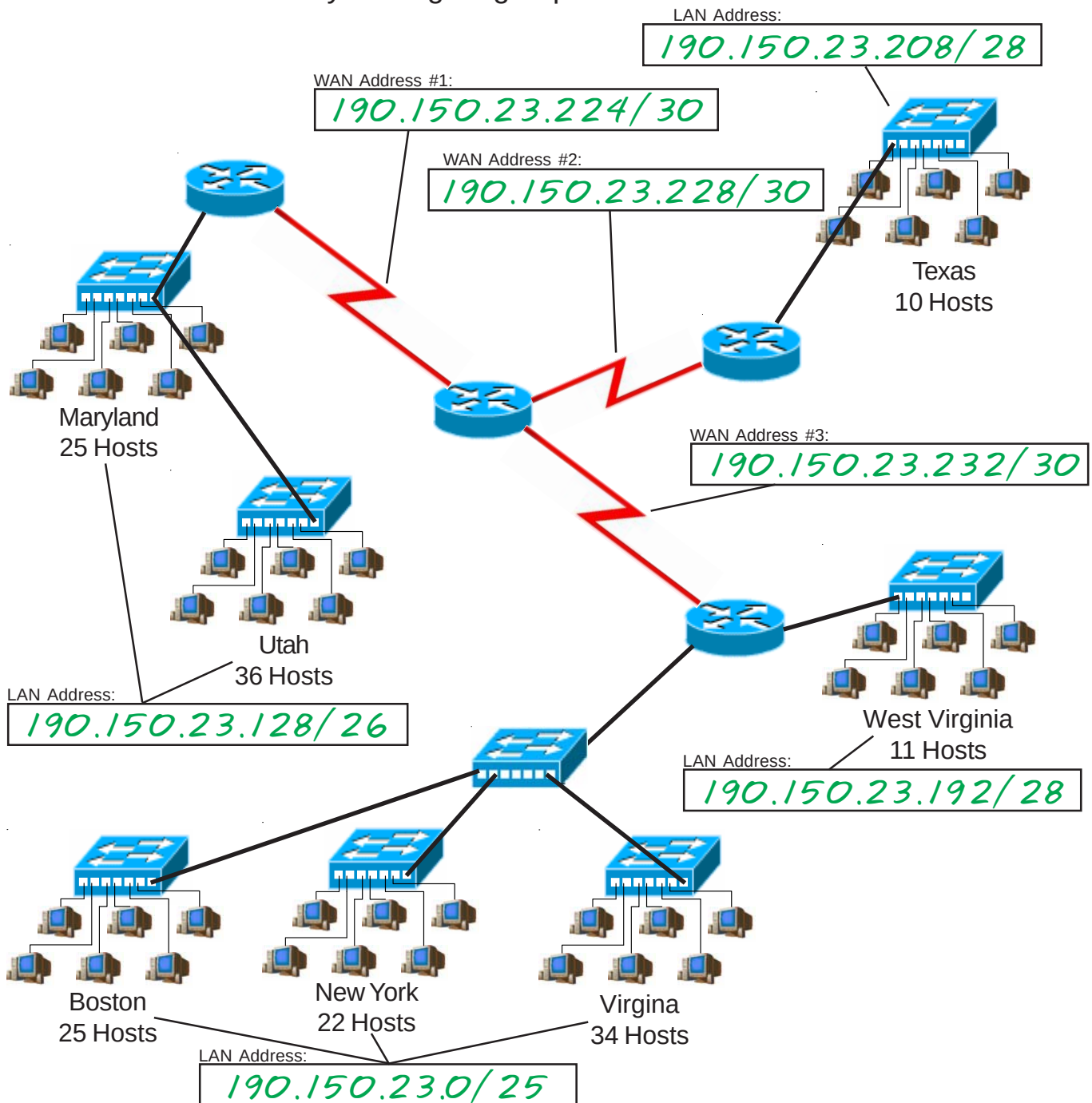
/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	16-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM Addressing

VLSM Chart Method

Problem 23

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 190.150.23.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

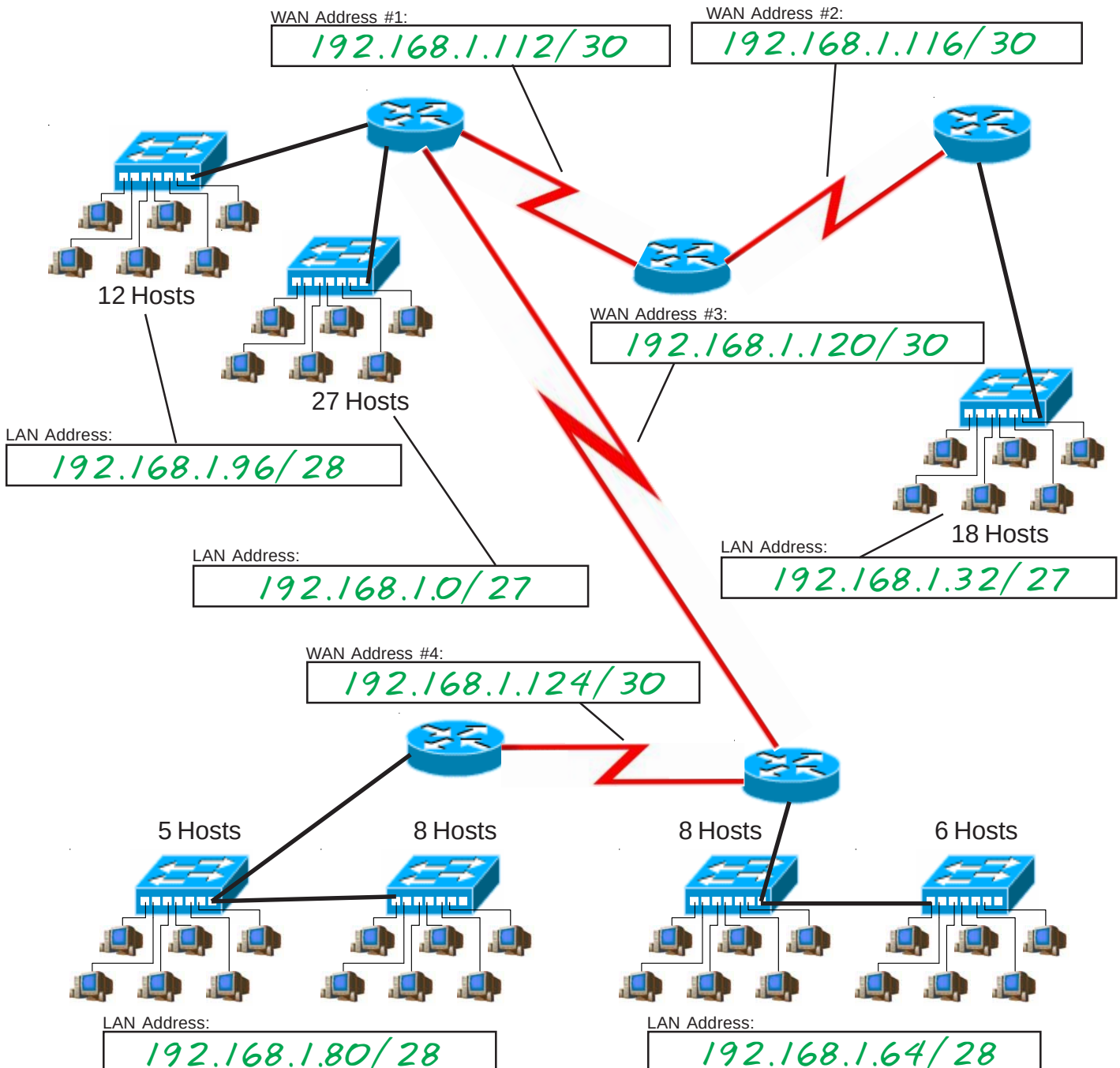
/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	16-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

VLSM Addressing

VLSM Chart Method

Problem 24

Using the network diagram and information given create an addressing scheme which utilizes variable-length subnet masks. Show the subnet address and CIDR in the boxes below, color or shade the sub-subnets used in the chart. This company will be using the class C address 192.168.1.0. Remember to start with your largest groups first.



Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

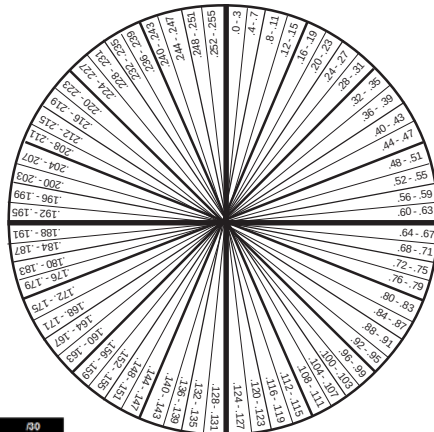
/24	/25	/26	/27	/28	/29	/30		
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts		
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3		
						4-7		
				16-31	8-15	8-11		
						12-15		
					32-47	16-23	16-19	
							20-23	
						48-63	24-31	24-27
								28-31
			64-127	32-63	32-39	32-35		
						36-39		
					40-47	40-43		
						44-47		
				48-63	48-55	48-51		
						52-55		
					56-63	56-59		
							60-63	
		64-127	64-95	64-79	64-71	64-67		
						68-71		
					72-79	72-75		
						76-79		
				80-95	80-87	80-83		
						84-87		
					88-95	88-91		
							92-95	
			96-127	96-111	96-103	96-99		
						100-103		
					104-111	104-107		
						108-111		
				112-127	112-119	112-115		
						116-119		
						120-123		
						124-127		
	128-255	128-191	128-159	128-143	128-135	128-131		
						132-135		
				144-159	136-143	136-139		
						140-143		
					152-159	144-151	144-147	
							148-151	
					160-191	160-175	152-155	152-155
								156-159
			168-175	160-163				
				164-167				
			176-191	168-171		168-171		
						172-175		
				184-191		176-179	176-179	
							180-183	
			192-255	192-223	192-207	184-187	184-187	
							188-191	
		208-215				192-195		
						196-199		
		216-223			200-203			
						204-207		
					224-231	208-211		
						212-215		
		224-255		224-239	216-219	216-219		
						220-223		
					232-239	224-227		
						228-231		
				240-255	232-235	232-235		
						236-239		
					248-255	240-243		
						244-247		
			248-251					
			252-255					

Practical VLSM

Problems

Use the VLSM method of your choice to complete the following problems.

0	8	32	40	128	136	160	168
	3	11	35	43	131	139	163
4	12	36	44	132	140	164	172
	7	15	39	47	135	143	167
16	24	48	56	144	152	176	184
	19	27	51	59	147	155	179
20	28	52	60	148	156	180	188
	23	31	55	63	151	159	183
64	72	96	104	192	200	224	232
	67	75	99	107	195	203	227
68	76	100	108	196	204	228	236
	71	79	103	111	199	207	231
80	88	112	120	208	216	240	248
	83	91	115	123	211	219	243
84	92	116	124	212	220	244	252
	87	95	119	127	215	223	247
							255



VLSM Chart 24-30 Bits (4th octet)

24	25	26	27	28	29	30
256 hosts	128 hosts	64 hosts	32 hosts	16 hosts	8 hosts	4 hosts
0-255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
						8-15
						16-23
						24-31
						32-39
						40-47
						48-55
						56-63
						64-71
						72-79
						80-87
						88-95
						96-103
						104-111
						112-119
						120-127
						128-135
						136-143
						144-151
						152-159
						160-167
						168-175
						176-183
						184-191
						192-199
						200-207
						208-215
						216-223
						224-231
						232-239
						240-247
						248-255

VLSM Addressing

(Sample)

Problem 25

You are developing a school network with the class C address 192.168.2.0/24. There will be three computer labs with 30 computers each that need to be on different sub-subnets. Forty eight classrooms with one computer each that will comprise a single sub-subnet. The administrative office and guidance office contain a total of seven computers which will need to be grouped together. Plan for four more mini labs with six computers to each sub-subnetwork. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.2.0	/26	192.168.2.1	192.168.2.62	192.168.2.63
2	192.168.2.64	/27	192.168.2.65	192.168.2.94	192.168.2.95
3	192.168.2.96	/27	192.168.2.97	192.168.2.126	192.168.2.127
4	192.168.2.128	/27	192.168.2.129	192.168.2.158	192.168.2.159
5	192.168.2.160	/28	192.168.2.161	192.168.2.174	192.168.2.175
6	192.168.2.176	/29	192.168.2.177	192.168.2.182	192.168.2.183
7	192.168.2.184	/29	192.168.2.185	192.168.2.190	192.168.2.191
8	192.168.2.192	/29	192.168.2.193	192.168.2.198	192.168.2.199
9	192.168.2.200	/29	192.168.2.201	192.168.2.206	192.168.2.207
10					
11					
12					
13					
14					

VLSM Addressing

(Sample)

Problem 26

You are setting up a small business network with the class C address 220.55.80.0/24. The marketing division will need 12 computers. Research and development needs 27 computers. The reception area will need two computers. Management requires 19 computers. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	220.55.80.0	/ 27	220.55.80.1	220.55.80.30	220.55.80.31
2	220.55.80.32	/ 27	220.55.80.	220.55.80.62	220.55.80.63
3	220.55.80.64	/ 28	220.55.80.65	220.55.80.78	220.55.80.79
4	220.55.80.80	/ 30	220.55.80.81	220.55.80.82	220.55.80.83
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 27

You are setting up a medium sized network with the class C address 222.37.34.0/24. Marketing needs 29 computers. Research and development needs 110 computers. Bookkeeping will use 12 computers. The reception area will need three computers. Management requires 60 computers. Divide the network using variable length subnet masks. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	222.37.34.0	/25	222.37.34.1	222.37.34.126	222.37.34.127
2	222.37.34.128	/26	222.37.34.129	222.37.34.190	222.37.34.191
3	222.37.34.192	/27	222.37.34.193	222.37.34.222	222.37.34.223
4	222.37.34.224	/28	222.37.34.225	222.37.34.238	222.37.34.239
5	222.37.34.240	/29	222.37.34.241	222.37.34.246	222.37.34.247
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 28

A shipping company needs to set up its network across several locations. The Denver office needs six computers. The Waco office needs 22 computers. The Fargo office will need five computers. The WAN links between all three locations need to be included in the solution. Using the IP address 192.168.10.0/24 divide the network using VLSM. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.10.0	/27	192.168.10.1	192.168.10.30	192.168.10.31
2	192.168.10.32	/29	192.168.10.33	192.168.10.38	192.168.10.39
3	192.168.10.40	/29	192.168.10.41	192.168.10.46	192.168.10.47
4	192.168.10.48	/30	192.168.10.49	192.168.10.50	192.168.10.51
5	192.168.10.52	/30	192.168.10.53	192.168.10.54	192.168.10.55
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 29

A new school is being built in the local school district. It will have three computer labs with 28 computers each. There will be 58 classrooms with 2 computers each that need to be on one sub-subnet. The office staff and administrators will need 7 computers. The guidance and attendance office will have 5 computers. The school has been given the address 223.145.75.0/24. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	223.145.75.0	/25	223.145.75.1	223.145.75.126	223.145.75.127
2	223.145.75.128	/27	223.145.75.129	223.145.75.158	223.145.75.159
3	223.145.75.160	/27	223.145.75.161	223.145.75.190	223.145.75.191
4	223.145.75.192	/27	223.145.75.193	223.145.75.222	223.145.75.223
5	223.145.75.224	/28	223.145.75.225	223.145.75.238	223.145.75.239
6	223.145.75.240	/29	223.145.75.241	223.145.75.246	223.145.75.247
7	223.145.75.248	/29	223.145.75.249	223.145.75.254	223.145.75.255
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 30

A local college is setting up a campus wide network. The technology wing will be on its own network address of 192.168.250.0/24. The office wing will include 15 computers. There are 2 labs of 20 computers each, 2 labs of 30 computers each and one lab of 35 computers. Complete the information required below. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.250.0	/26	92.168.250.1	92.168.250.62	92.168.250.63
2	92.168.250.64	/27	92.168.250.65	92.168.250.94	92.168.250.95
3	92.168.250.96	/27	92.168.250.97	92.168.250.126	92.168.250.127
4	92.168.250.128	/27	92.168.250.129	92.168.250.158	92.168.250.159
5	92.168.250.160	/27	92.168.250.161	92.168.250.190	92.168.250.191
6	92.168.250.192	/27	92.168.250.193	92.168.250.222	92.168.250.223
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 31

You are setting up a network for a company in four locations. Location A has 8 computers. Location B has 122 computers. Location C has 4 computers. Location D has 55 computers. There is a WAN connection between all four locations. Complete the information required below using the class C address 192.168.10.0. Remember to work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.10.0	/25	192.168.10.1	192.168.10.126	192.168.10.127
2	192.168.10.128	/26	192.168.10.129	192.168.10.190	192.168.10.191
3	192.168.10.192	/28	192.168.10.193	192.168.10.206	192.168.10.207
4	192.168.10.208	/29	192.168.10.209	192.168.10.214	192.168.10.215
5	192.168.10.216	/30	192.168.10.217	192.168.10.218	192.168.10.219
6	192.168.10.220	/30	192.168.10.221	192.168.10.222	192.168.10.223
7	192.168.10.224	/30	192.168.10.225	192.168.10.226	192.168.10.227
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

Problem 32

A college dormitory is being remodeled. A new network is being installed. There are 50 dorm rooms with two drops each that will be on one sub-subnet. The offices will have 5 drops. The reception desk will have three drops. A small study hall will include 30 drops. Using the IP address 192.168.12.0/24 complete the information required below using VLSM. Work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (X)	First Usable Host	Last Usable Host	Broadcast Address
1	192.168.12.0	/25	192.168.12.1	192.168.12.126	192.168.12.127
2	192.168.12.128	/27	192.168.12.129	192.168.12.158	192.168.12.159
3	192.168.12.160	/29	192.168.12.161	192.168.12.166	192.168.12.167
4	192.168.12.168	/29	192.168.12.169	192.168.12.174	192.168.12.175
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

VLSM Addressing

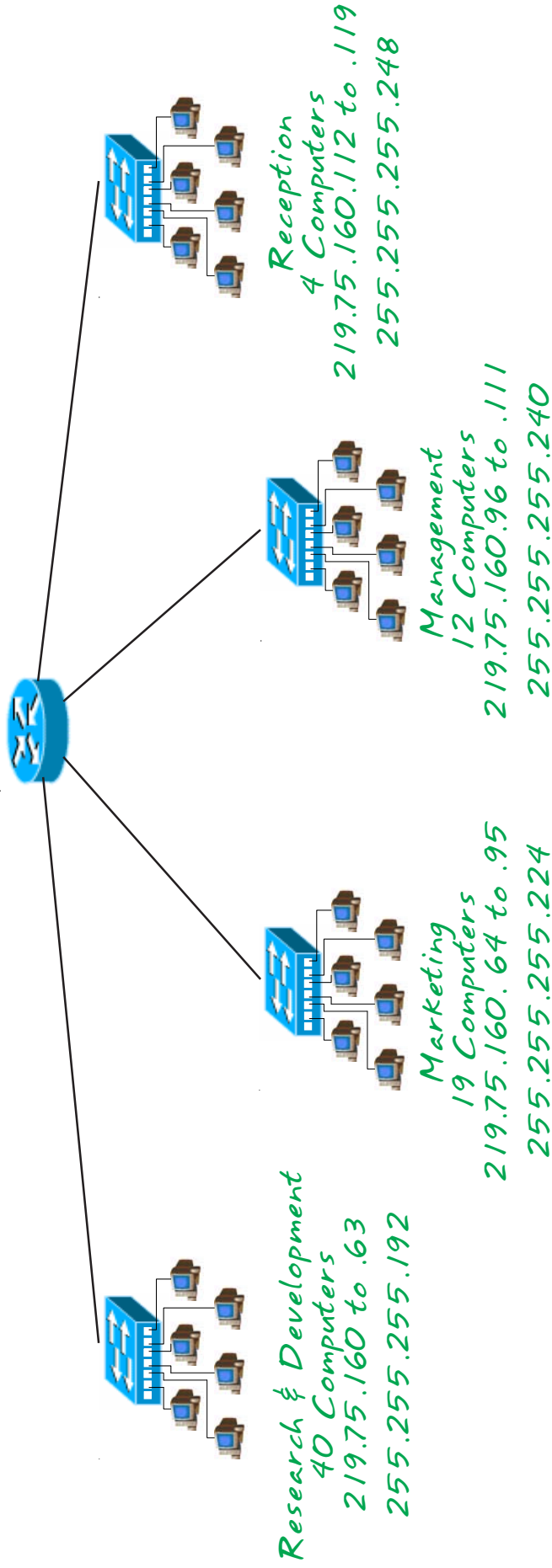
Problem 33

You are setting up a business network with the class C address 219.75.160.0/24. The marketing division will need 19 computers. Research and development needs 40 computers. The reception area will need four computers. Management requires 12 computers. Divide the network using variable length subnet information. On the opposite page draw a detailed map of this network. Include the name and sub-subnet IP addresses for each branch of the network with the subnet mask. One router with four ethernet ports will be used for this network.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	219.75.160.0	/26	219.75.160.1	219.75.160.62	219.75.160.63
2	219.75.160.64	/27	219.75.160.65	219.75.160.94	219.75.160.95
3	219.75.160.96	/28	219.75.160.97	219.75.160.110	219.75.160.111
4	219.75.160.112	/29	219.75.160.113	219.75.160.118	219.75.160.119
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					

Problem 33 - Detailed Map

Draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network, and the subnet mask.



VLSM Addressing

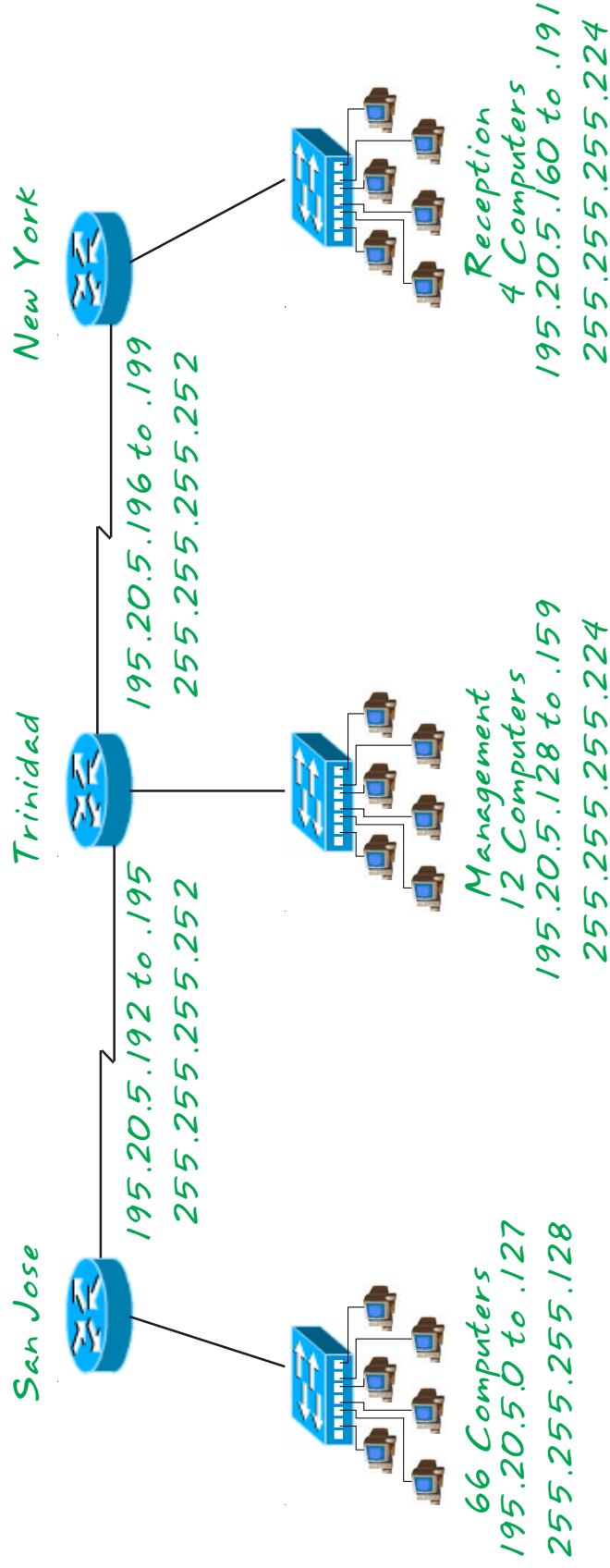
Problem 34

A small company needs to set up its network across several locations. The New York branch office needs 15 computers. The San Jose office needs 66 computers. The Trinidad office will need 18 computers. You will need two WAN links between the routers. Using the IP address 195.20.5.0/24 divide the network using VLSM. On the opposite page draw a detailed map of this network. Include the name and subnet IP addresses information for each branch of the network. Label the WAN links with the same information. Complete the information required below. Work from largest to smallest.

Subnet	Subnet Address	Subnet Mask (/X)	First Usable Host	Last Usable Host	Broadcast Address
1	195.20.5.0	/25	195.20.5.1	195.20.5.126	195.20.5.127
2	195.20.5.128	/27	195.20.5.129	195.20.5.158	195.20.5.159
3	195.20.5.160	/27	195.20.5.161	195.20.5.190	195.20.5.191
4	195.20.5.192	/30	195.20.5.193	195.20.5.194	195.20.5.195
5	195.20.5.196	/30	195.20.5.197	195.20.5.198	195.20.5.199
6					
7					
8					
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11					
12					
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14					

Problem 34 - Detailed Map

Draw a detailed map of this network. Include the name and sub-subnet IP addresses information for each branch of the network.



10 . 0 . 0 . 0

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

04	05	06	07	08	09	10
200-205	206-210	211-215	216-220	221-225	226-230	231-235
0-01	0-02	0-03	0-04	0-05	0-06	0-07
0-08	0-09	0-10	0-11	0-12	0-13	0-14
0-15	0-16	0-17	0-18	0-19	0-20	0-21
0-22	0-23	0-24	0-25	0-26	0-27	0-28
0-29	0-30	0-31	0-32	0-33	0-34	0-35
0-36	0-37	0-38	0-39	0-40	0-41	0-42
0-43	0-44	0-45	0-46	0-47	0-48	0-49
0-50	0-51	0-52	0-53	0-54	0-55	0-56
0-57	0-58	0-59	0-60	0-61	0-62	0-63
0-64	0-65	0-66	0-67	0-68	0-69	0-70
0-71	0-72	0-73	0-74	0-75	0-76	0-77
0-78	0-79	0-80	0-81	0-82	0-83	0-84
0-85	0-86	0-87	0-88	0-89	0-90	0-91
0-92	0-93	0-94	0-95	0-96	0-97	0-98
0-99	0-100	0-101	0-102	0-103	0-104	0-105
0-106	0-107	0-108	0-109	0-110	0-111	0-112
0-113	0-114	0-115	0-116	0-117	0-118	0-119
0-120	0-121	0-122	0-123	0-124	0-125	0-126
0-127	0-128	0-129	0-130	0-131	0-132	0-133
0-134	0-135	0-136	0-137	0-138	0-139	0-140
0-141	0-142	0-143	0-144	0-145	0-146	0-147
0-148	0-149	0-150	0-151	0-152	0-153	0-154
0-155	0-156	0-157	0-158	0-159	0-160	0-161
0-162	0-163	0-164	0-165	0-166	0-167	0-168
0-169	0-170	0-171	0-172	0-173	0-174	0-175
0-176	0-177	0-178	0-179	0-180	0-181	0-182
0-183	0-184	0-185	0-186	0-187	0-188	0-189
0-190	0-191	0-192	0-193	0-194	0-195	0-196
0-197	0-198	0-199	0-200	0-201	0-202	0-203
0-204	0-205	0-206	0-207	0-208	0-209	0-210
0-211	0-212	0-213	0-214	0-215	0-216	0-217
0-218	0-219	0-220	0-221	0-222	0-223	0-224
0-225	0-226	0-227	0-228	0-229	0-230	0-231
0-232	0-233	0-234	0-235	0-236	0-237	0-238
0-239	0-240	0-241	0-242	0-243	0-244	0-245
0-246	0-247	0-248	0-249	0-250	0-251	0-252
0-253	0-254	0-255	0-256	0-257	0-258	0-259
0-260	0-261	0-262	0-263	0-264	0-265	0-266
0-267	0-268	0-269	0-270	0-271	0-272	0-273
0-274	0-275	0-276	0-277	0-278	0-279	0-280
0-281	0-282	0-283	0-284	0-285	0-286	0-287
0-288	0-289	0-290	0-291	0-292	0-293	0-294
0-295	0-296	0-297	0-298	0-299	0-300	0-301
0-302	0-303	0-304	0-305	0-306	0-307	0-308
0-309	0-310	0-311	0-312	0-313	0-314	0-315
0-316	0-317	0-318	0-319	0-320	0-321	0-322
0-323	0-324	0-325	0-326	0-327	0-328	0-329
0-330	0-331	0-332	0-333	0-334	0-335	0-336
0-337	0-338	0-339	0-340	0-341	0-342	0-343
0-344	0-345	0-346	0-347	0-348	0-349	0-350
0-351	0-352	0-353	0-354	0-355	0-356	0-357
0-358	0-359	0-360	0-361	0-362	0-363	0-364
0-365	0-366	0-367	0-368	0-369	0-370	0-371
0						

VLSM

with

Class A and B Addresses

We've gone over the practical applications of using VLSM on class C addresses. The same approach works with class A and B addresses. For example an ISP may have a class A address which it needs to subnet between its customers. Each customer may need to take their addresses and subnet them again in order to use them more effectively. The real trick to this is to remember which octet of the IP address you are working with.

Sample Problem 35

Part 1 of 3

Use the **Class A** address chart to break down the address for different ISP customers. At this stage of the problem you are creating subnets using the second octet of the IP address.

ISP Addresses 15.0.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Customer #1	8 million	<i>15.0.0.0 to 15.127.255.255</i>	<i>/9</i>
Customer #2	2 million	<i>15.128.0.0 to 15.159.255.255</i>	<i>/11</i>
Customer #3	2,000,000	<i>15.160.0.0 to 15.191.255.255</i>	<i>/11</i>
Customer #4	1,000,000	<i>15.192.0.0 to 15.207.255.255</i>	<i>/12</i>
Customer #5	500,000	<i>15.208.0.0 to 15.215.255.255</i>	<i>/13</i>
Customer #6	450,000	<i>15.216.0.0 to 15.223.255.255</i>	<i>/13</i>
Customer #7	200,000	<i>15.224.0.0 to 15.227.255.255</i>	<i>/14</i>
Customer #8	130,000	<i>15.228.0.0 to 15.229.255.255</i>	<i>/15</i>
Customer #9	100,000	<i>15.230.0.0 to 15.231.255.255</i>	<i>/15</i>

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
			16-31	16-23		16-19	8-9
						20-23	10-11
					24-31	24-27	12-13
						28-31	14-15
			32-63	32-39		32-35	16-17
						36-39	18-19
				40-47		40-43	20-21
						44-47	22-23
				48-55		48-51	24-25
						52-55	26-27
				56-63		56-59	28-29
						60-63	30-31
		64-127	64-95	64-71		64-67	32-33
						68-71	34-35
				72-79		72-75	36-37
						76-79	38-39
			80-95	80-87		80-83	40-41
						84-87	42-43
				88-95		88-91	44-45
						92-95	46-47
			96-127	96-103		96-99	48-49
						100-103	50-51
				104-111		104-107	52-53
						108-111	54-55
				112-119		112-115	56-57
						116-119	58-59
			120-127			120-123	60-61
						124-127	62-63
	128-255	128-191	128-159	128-143		128-131	64-65
						132-135	66-67
				136-143		136-139	68-69
						140-143	70-71
			144-159	144-151		144-147	72-73
						148-151	74-75
				152-159		152-155	76-77
						156-159	78-79
			160-191	160-167		160-163	80-81
						164-167	82-83
				168-175		168-171	84-85
						172-175	86-87
				176-183		176-179	88-89
						180-183	90-91
				184-191		184-187	92-93
						188-191	94-95
		192-255	192-223	192-199		192-195	96-97
						196-199	98-99
				200-207		200-203	100-101
						204-207	102-103
			208-223	208-215		208-211	104-105
						212-215	106-107
				216-223		216-219	108-109
						220-223	110-111
		224-255	224-239	224-231		224-227	112-113
						228-231	114-115
				232-239		232-235	116-117
						236-239	118-119
			240-255	240-247		240-243	120-121
						244-247	122-123
				248-255		248-251	124-125
						252-255	126-127

VLSM

with

Class A and B Addresses

Sample Problem 35

Part 2 of 3

Customer #5 has a total of 524,288 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for their different clients. At this stage of the problem you are creating sub-subnets with the third octet of the IP address.

ISP Addresses 15.208.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Client #1	7,500	<i>15.208.0.0 to 15.208.31.255</i>	<i>/19</i>
Client #2	5,000	<i>15.208.32.0 to 15.208.63.255</i>	<i>/19</i>
Client #3	4,500	<i>15.208.64.0 to 15.208.95.255</i>	<i>/19</i>
Client #4	2,000	<i>15.208.96.0 to 15.208.103.255</i>	<i>/21</i>
Client #5	1,450	<i>15.208.104.0 to 15.208.111.255</i>	<i>/21</i>
Client #6	1,150	<i>15.208.112.0 to 15.208.119.255</i>	<i>/21</i>
Client #7	900	<i>15.208.120.0 to 15.208.123.255</i>	<i>/22</i>
Client #8	750	<i>15.208.124.0 to 15.208.127.255</i>	<i>/22</i>
Client #9	450	<i>15.208.128.0 to 15.208.129.255</i>	<i>/23</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
						16-19	8-9
				16-31	16-23	16-19	10-11
					20-23	20-23	12-13
					24-31	24-27	14-15
						28-31	16-17
						32-35	18-19
		32-63	32-63	32-47	32-39	36-39	20-21
					40-47	40-43	22-23
					48-55	48-51	24-25
					56-63	56-59	26-27
						60-63	28-29
				48-63			30-31
							32-33
							34-35
							36-37
							38-39
		64-127	64-95	64-79	64-71	64-67	40-41
					72-79	68-71	42-43
						72-75	44-45
						76-79	46-47
				80-95	80-87	80-83	48-49
					88-95	88-91	50-51
						92-95	52-53
				96-111	96-103	96-99	54-55
					104-111	104-107	56-57
						108-111	58-59
					112-119	112-115	60-61
			96-127	112-127		116-119	62-63
							64-65
							66-67
							68-69
							70-71
							72-73
							74-75
							76-77
							78-79
							80-81
	128-255	128-191	128-159	128-143	128-135	128-131	82-83
					136-143	132-135	84-85
						136-139	86-87
						140-143	88-89
				144-159	144-151	144-147	90-91
					152-159	148-151	92-93
						152-155	94-95
			160-191	160-175	160-167	156-159	96-97
					168-175	164-167	98-99
						168-171	100-101
						172-175	102-103
				176-191	176-183	176-179	104-105
					180-183	180-183	106-107
					184-191	184-187	108-109
						188-191	110-111
			192-255	192-207	192-199	192-195	112-113
					200-207	196-199	114-115
						200-203	116-117
				208-223	208-215	204-207	118-119
					212-215	208-211	120-121
					216-223	212-215	122-123
						216-219	124-125
						220-223	126-127
				224-239	224-231	224-227	128-129
					232-239	228-231	130-131
						232-235	132-133
			224-255	240-247	236-239	236-239	134-135
					240-243	240-243	136-137
				248-255	244-247	244-247	138-139
					248-251	248-251	140-141
					252-255	252-255	142-143
							144-145
							146-147
							148-149

VLSM

with

Class A and B Addresses

Sample Problem 35

Part 3 of 3

Client #8 has a total of 1,024 addresses. Use the **Class C** address chart to break down the sub-subnetwork addresses for their different branch offices. At this stage of the problem you are creating sub-subnets with the fourth octet of the IP address.

ISP Addresses 15.208.124.**0**

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Branch #1	100	<i>15.208.124.0 to 15.208.124.127</i>	<i>/ 25</i>
Branch #2	55	<i>15.208.124.128 to 15.208.124.191</i>	<i>/ 26</i>
Branch #3	25	<i>15.208.124.192 to 15.208.124.223</i>	<i>/ 27</i>
Branch #4	6	<i>15.208.124.224 to 15.208.124.231</i>	<i>/ 29</i>
Branch #5	4	<i>15.208.124.232 to 15.208.124.239</i>	<i>/ 29</i>
Branch #6	2	<i>15.208.124.240 to 15.208.124.243</i>	<i>/ 30</i>
Branch #7	2	<i>15.208.124.244 to 15.208.124.247</i>	<i>/ 30</i>
Branch #8	2	<i>15.208.124.248 to 15.208.124.251</i>	<i>/ 30</i>
Branch #9	2	<i>15.208.124.252 to 15.208.124.255</i>	<i>/ 30</i>

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

VLSM

with

Class A and B Addresses

Problem 36

Part 1 of 3

The school system you are working for is using the private address of 172.32.0.0 to subnet the entire district. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different schools and offices.

At this stage of the problem you are creating sub-subnets with the third octet of the IP address. Remember which octet of the IP address you are working in.

School System Address 172.32.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
North High	2,400	<i>172.32.0.0 to 172.32.15.255</i>	<i>/20</i>
South High	2,000	<i>172.32.16.0 to 172.32.23.255</i>	<i>/21</i>
North Middle	1,200	<i>172.32.24.0 to 172.32.31.255</i>	<i>/21</i>
South Middle	1,000	<i>172.32.32.0 to 172.32.35.255</i>	<i>/22</i>
Central Elem.	550	<i>172.32.36.0 to 172.32.39.255</i>	<i>/22</i>
Southern Elem.	475	<i>172.32.40.0 to 172.32.41.255</i>	<i>/23</i>
Eastern Elem.	450	<i>172.32.42.0 to 172.32.43.255</i>	<i>/23</i>
Central Office	400	<i>172.32.44.0 to 172.32.45.255</i>	<i>/23</i>
Western Elem.	300	<i>172.32.46.0 to 172.32.47.255</i>	<i>/23</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	4-5
					8-15	8-11	6-7
						12-15	8-9
				16-31	16-23	16-19	10-11
						20-23	12-13
					24-31	24-27	14-15
						28-31	16-17
			32-63	32-47	32-39	32-35	18-19
						36-39	20-21
					40-47	40-43	22-23
						44-47	24-25
				48-63	48-55	48-51	26-27
						52-55	28-29
					56-63	56-59	30-31
						60-63	32-33
		64-127	64-95	64-79	64-71	64-67	34-35
						68-71	36-37
					72-79	72-75	38-39
						76-79	40-41
			80-95	80-87	80-83	80-81	42-43
						84-87	44-45
					88-95	88-91	46-47
						92-95	48-49
			96-127	96-103	96-99	96-97	50-51
						100-103	52-53
				104-111	104-107	104-105	54-55
						108-111	56-57
				112-127	112-119	112-115	58-59
						116-119	60-61
					120-127	120-123	62-63
						124-127	64-65
	128-255	128-191	128-159	128-143	128-135	128-131	66-67
						132-135	68-69
					136-143	136-139	70-71
						140-143	72-73
			144-159	144-151	144-147	144-145	74-75
						148-151	76-77
					152-159	152-155	78-79
						156-159	80-81
			160-191	160-175	160-167	160-163	82-83
						164-167	84-85
					168-175	168-171	86-87
						172-175	88-89
			176-191	176-183	176-179	176-177	90-91
						180-183	92-93
					184-191	184-187	94-95
						188-191	96-97
		192-255	192-223	192-207	192-199	192-195	98-99
						196-199	100-101
					200-207	200-203	102-103
						204-207	104-105
			208-223	208-215	208-211	208-209	106-107
						212-215	108-109
					216-223	216-219	110-111
						220-223	112-113
			224-255	224-231	224-227	224-225	114-115
						228-231	116-117
				232-239	232-235	232-233	118-119
						236-239	120-121
				240-255	240-243	240-241	122-123
						244-247	124-125
					248-251	248-249	126-127
					252-255	252-253	128-129

VLSM

with

Class A and B Addresses

Problem 36

Part 2 of 3

Eastern Elementary has been given 512 hosts, with the address range of 172.32.42.0 / 21 (255.255.248.0).

Based on the information below supply the required address ranges and subnet masks for each school area. Use the **Class C** address chart to break down the sub-subnetworks.

Hint:

Another way to look at this problem is to see that with the third octet range of 42 to 43 you have access to 2 groups of 255 addresses (172.32.42.0 and 172.32.43.0). Think in terms of having two Class C VLSM charts.

Eastern Elementary School
Address Range 172.32.42.0 to 172.32.43.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Students	250	<i>172.32.42.0 to 172.32.42.255</i>	<i>/24</i>
Printers	45	<i>172.32.43.0 to 172.32.43.63</i>	<i>/26</i>
Staff	40	<i>172.32.43.64 to 172.32.43.127</i>	<i>/26</i>
Network Devices	25	<i>172.32.43.128 to 172.32.43.159</i>	<i>/27</i>
Administrative	12	<i>172.32.43.160 to 172.32.43.175</i>	<i>/28</i>

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	160-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM

with

Class A and B Addresses

Problem 36

Part 3 of 3

South High in part 1 of this problem has been given 2,048 hosts, with the address range of 172.32.16.0 / 21 (255.255.248.0).

Based on the information below supply the required address ranges and subnet masks for each school area. Use both the **Class B** and **Class C** address charts to break down the sub-subnetwork addresses for the different areas of the network.

Hint:

With this problem you are creating sub-subnets with both the third and fourth octets of the IP address. You may need to use the Class B VLSM chart for the *Students* addressing information. All the other addresses will be using the Class C VLSM chart. Another way to look at this problem is to see that with the third octet range of 16 to 23 you have access to 8 groups of 255 addresses or eight Class C VLSM charts.

South High School
Address Range 172.32.16.0 to 172.32.23.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Students	1,000	172.32.16.0 to 172.32.19.255	/22
Network Devices	250	172.32.20.0 to 172.32.20.255	/24
Printers	200	172.32.21.0 to 172.32.21.255	/24
Staff	150	172.32.22.0 to 172.32.22.255	/24
Administrative	50	172.32.23.0 to 172.32.23.63	/26

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24		/25		/26		/27		/28		/29		/30	
255.255.255.0 256 Hosts		255.255.255.128 128 Hosts		255.255.255.192 64 Hosts		255.255.255.224 32 Hosts		255.255.255.240 16 Hosts		255.255.255.248 8 Hosts		255.255.255.252 4 Hosts	
0 - 255				0-127	0-63	0-31	0-15	0-7		0-3		4-7	
								8-15		8-11		12-15	
								16-23		16-19		20-23	
								24-31		24-27		28-31	
							16-31	32-39		32-35		36-39	
								40-47		40-43		44-47	
								48-55		48-51		52-55	
								56-63		56-59		60-63	
						32-63		64-71		64-67		68-71	
								72-79		72-75		76-79	
								80-87		80-83		84-87	
								88-95		88-91		92-95	
							64-127	96-103		96-99		100-103	
								104-111		104-107		108-111	
								112-119		112-115		116-119	
								120-127		120-123		124-127	
					64-95	128-135		128-131		132-135			
						136-143		136-139		140-143			
						144-151		144-147		148-151			
						152-159		152-155		156-159			
						96-127	160-167		160-163		164-167		
							168-175		168-171		172-175		
							176-183		176-179		180-183		
							184-191		184-187		188-191		
					128-191		192-199		192-195		196-199		
							200-207		200-203		204-207		
							208-215		208-211		212-215		
							216-223		216-219		220-223		
						128-255	224-231		224-227		228-231		
							232-239		232-235		236-239		
							240-247		240-243		244-247		
							248-255		248-251		252-255		

VLSM

with

Class A and B Addresses

Problem 37

Part 1 of 3

The company you are working for is using the IP address 110.0.0.0 sub-subneted for multiple offices around the world. Use the **Class A** address chart to break down the sub-subnetwork addresses for the different offices.

At this stage of the problem you are creating sub-subnets with the third octet of the IP address. Remember which octet of the IP address you are working in.

Company Address 110.0.0.0

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Moskva	3,050,000	<i>110.0.0.0 to 110.63.255.255</i>	<i>/10</i>
New York	1,540,000	<i>110.64.0.0 to 110.95.255.255</i>	<i>/11</i>
St. Petersburg	1,075,000	<i>110.96.0.0 to 110.127.255.255</i>	<i>/11</i>
London	975,000	<i>110.128.0.0 to 110.143.255.255</i>	<i>/12</i>
Ekaterinoburg	525,000	<i>110.144.0.0 to 110.159.255.255</i>	<i>/12</i>
Munchen	450,000	<i>110.160.0.0 to 110.167.255.255</i>	<i>/13</i>
Napoli	150,000	<i>110.168.0.0 to 110.171.255.255</i>	<i>/14</i>
Birmingham	130,000	<i>110.172.0.0 to 110.173.255.255</i>	<i>/15</i>
Rotterdam	95,000	<i>110.174.0.0 to 110.175.255.255</i>	<i>/15</i>

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
					16-31	8-11	4-5
						12-15	6-7
			32-63	32-47	16-23	16-19	8-9
					20-23	18-19	10-11
				48-63	24-31	20-23	12-13
					32-39	24-27	14-15
				56-63	40-47	28-31	16-17
					44-47	30-31	18-19
		64-127	64-95	64-79	32-35	32-35	20-21
					36-39	36-39	22-23
				80-95	40-43	40-43	24-25
					44-47	44-47	26-27
			96-127	96-111	48-51	48-51	28-29
					52-55	52-55	30-31
				112-127	56-59	56-59	32-33
					60-63	60-63	34-35
	128-255	128-191	128-159	128-143	64-71	64-67	36-37
					72-79	68-71	38-39
				144-159	80-87	80-83	40-41
					88-95	84-87	42-43
			160-191	160-175	96-99	88-91	44-45
					100-103	92-95	46-47
				176-191	104-111	96-99	48-49
					112-119	100-103	50-51
		192-255	192-223	192-207	116-119	104-107	52-53
					120-123	108-111	54-55
				208-223	124-127	112-115	56-57
					128-135	116-119	58-59
			224-255	224-239	132-135	120-123	60-61
					136-139	124-127	62-63
				240-255	140-143	128-131	64-65
					144-147	132-135	66-67
				248-255	148-151	136-139	68-69
					152-155	140-143	70-71
				252-255	156-159	144-147	72-73
					160-163	148-149	74-75
					164-167	152-153	76-77
					168-171	156-157	78-79
					172-175	160-161	80-81
					176-179	164-165	82-83
					180-183	168-169	84-85
					184-187	172-173	86-87
					188-191	176-177	88-89
					192-195	180-181	90-91
					196-199	184-185	92-93
					200-203	188-189	94-95
					204-207	192-193	96-97
					208-211	196-197	98-99
					212-215	200-201	100-101
					216-219	204-205	102-103
					220-223	208-209	104-105
					224-227	212-213	106-107
					228-231	216-217	108-109
					232-235	220-221	110-111
					236-239	224-225	112-113
					240-243	228-229	114-115
					244-247	232-233	116-117
					248-251	236-237	118-119
					252-255	240-241	120-121
						244-245	122-123
						248-249	124-125
						252-253	126-127
						254-255	128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Problem 37

Part 2 of 3

London in part 1 of this problem has been given 1,048,576 hosts, with the address range of 110.128.0.0 to 110.143.255.255 /12 (255.240.0.0).

Based on the information below supply the required address ranges and subnet masks for each office. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different areas of the network.

London
Address Range 110.128.0.0 to 110.143.255.255

Customer Name	Number of Addresses	Address Range (Include subnet & broadcast addresses)	CIDR
Office #1	6,450	<i>110.128.0.0 to 110.128.31.255</i>	<i>/19</i>
Office #2	3,780	<i>110.128.32.0 to 110.128.47.255</i>	<i>/19</i>
Office #3	2,750	<i>110.128.48.0 to 110.128.63.255</i>	<i>/19</i>
Office #4	2,000	<i>110.128.064.0 to 110.128.71.255</i>	<i>/21</i>
Office #5	1,000	<i>110.128.72.0 to 110.128.75.255</i>	<i>/21</i>
Office #6	845	<i>110.128.76.0 to 110.128.79.255</i>	<i>/21</i>
Office #7	500	<i>110.128.80.0 to 110.128.81.255</i>	<i>/22</i>
Office #8	450	<i>110.128.82.0 to 110.128.83.255</i>	<i>/22</i>
Office #9	300	<i>110.128.84.0 to 110.128.85.255</i>	<i>/23</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
						16-19	8-9
					16-23	20-23	10-11
						24-27	12-13
					24-31	28-31	14-15
			32-63	32-47	32-39	32-35	16-17
						36-39	18-19
					40-47	40-43	20-21
						44-47	22-23
				48-63	48-55	48-51	24-25
						52-55	26-27
					56-63	56-59	28-29
						60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
						68-71	34-35
					72-79	72-75	36-37
						76-79	38-39
				80-95	80-87	80-83	40-41
						84-87	42-43
					88-95	88-91	44-45
						92-95	46-47
			96-127	96-111	96-103	96-99	48-49
						100-103	50-51
					104-111	104-107	52-53
						108-111	54-55
				112-127	112-119	112-115	56-57
						116-119	58-59
					120-127	120-123	60-61
						124-127	62-63
	128-255	128-191	128-159	128-143	128-135	128-131	64-65
						132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
				144-159	144-151	144-147	72-73
						148-151	74-75
					152-159	152-155	76-77
						156-159	78-79
			160-191	160-175	160-167	160-163	80-81
						164-167	82-83
					168-175	168-171	84-85
						172-175	86-87
				176-191	176-183	176-179	88-89
						180-183	90-91
					184-191	184-187	92-93
						188-191	94-95
		192-255	192-223	192-207	192-199	192-195	96-97
						196-199	98-99
					200-207	200-203	100-101
						204-207	102-103
				208-223	208-215	208-211	104-105
						212-215	106-107
					216-223	216-219	108-109
						220-223	110-111
			224-255	224-239	224-231	224-227	112-113
						228-231	114-115
					232-239	232-235	116-117
						236-239	118-119
				240-255	240-247	240-243	120-121
						244-247	122-123
					248-255	248-251	124-125
						252-255	126-127

VLSM

with

Class A and B Addresses

Problem 37

Part 3 of 3

Office #7 in part 2 of this problem has been given 512 hosts, with the address range of 110.128.80.0 / 23 (255.255.254.0).

Based on the information below supply the required address ranges and subnet masks for each school area. Use the **Class C** address chart to break down the sub-subnetwork addresses for the different areas of the network. **Hint:** Another way to look at this problem is to see that with the third octet range of 80 to 81 you have access to 2 groups of 255 addresses or two Class C VLSM charts.

Office #7
Address Range 110.128.80.0 to 110.128.81.255

Customer Name	Number of Addresses	Address Range	CIDR
1st Floor	125	<i>110.128.80.0 to 110.128.80.127</i>	<i>/25</i>
2nd Floor	75	<i>110.128.80.128 to 110.128.80.255</i>	<i>/25</i>
5th Floor	50	<i>110.128.81.0 to 110.128.81.63</i>	<i>/26</i>
8th Floor	45	<i>110.128.81.64 to 110.128.81.127</i>	<i>/26</i>
4th Floor	30	<i>110.128.81.128 to 110.128.81.159</i>	<i>/27</i>
Basement	14	<i>110.128.81.160 to 110.128.81.175</i>	<i>/28</i>
7th Floor	12	<i>110.128.81.176 to 110.128.81.191</i>	<i>/28</i>
3rd Floor	6	<i>110.128.81.192 to 110.128.81.199</i>	<i>/29</i>
6th Floor	4	<i>110.128.81.200 to 110.128.81.207</i>	<i>/29</i>

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24 255.255.255.0 256 Hosts	/25 255.255.255.128 128 Hosts	/26 255.255.255.192 64 Hosts	/27 255.255.255.224 32 Hosts	/28 255.255.255.240 16 Hosts	/29 255.255.255.248 8 Hosts	/30 255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3 4-7
					8-15	8-11 12-15
				16-31	16-23	16-19 20-23
					24-31	24-27 28-31
			32-63	32-47	32-39	32-35 36-39
					40-47	40-43 44-47
				48-63	48-55	48-51 52-55
					56-63	56-59 60-63
		64-127	64-95	64-79	64-71	64-67 68-71
					72-79	72-75 76-79
				80-95	80-87	80-83 84-87
					88-95	88-91 92-95
			96-127	96-111	96-103	96-99 100-103
					104-111	104-107 108-111
				112-127	112-119	112-115 116-119
					120-127	120-123 124-127
	128-255	128-191	128-159	128-143	128-135	128-131 132-135
					136-143	136-139 140-143
				144-159	144-151	144-147 148-151
					152-159	152-155 156-159
			160-191	160-175	160-167	160-163 164-167
					168-175	168-171 172-175
				176-191	176-183	176-179 180-183
					184-191	184-187 188-191
		192-255	192-223	192-207	192-199	192-195 196-199
					200-207	200-203 204-207
				208-223	208-215	208-211 212-215
					216-223	216-219 220-223
			224-255	224-239	224-231	224-227 228-231
					232-239	232-235 236-239
				240-255	240-247	240-243 244-247
					248-255	248-251 252-255

VLSM

with

Class A and B Addresses

Problem 38

Part 1 of 4

Use the **Class A** address chart to break down the address for different business customers by country. At this stage of this problem you are creating subnets in the second octet of the IP address.

Addresses 75.0.0.0

Customer Name	Number of Addresses	Address Range	CIDR
United States	6.5 million	<i>75.0.0.0 to 75.127.255.255</i>	<i>/9</i>
China	4 million	<i>75.128.0.0 to 75.191.255.255</i>	<i>/10</i>
Japan	1 million	<i>75.192.0.0 to 75.207.255.255</i>	<i>/12</i>
Germany	500,000	<i>75.208.0.0 to 75.215.255.255</i>	<i>/13</i>
Russia	455,000	<i>75.216.0.0 to 75.223.255.255</i>	<i>/13</i>
Australia	450,000	<i>75.224.0.0 to 75.231.255.255</i>	<i>/13</i>
Brazil	125,000	<i>75.232.0.0 to 75.233.255.255</i>	<i>/15</i>
Canda	90,000	<i>75.234.0.0 to 75.235.255.255</i>	<i>/15</i>
Denmark	88,000	<i>75.236.0.0 to 75.237.255.255</i>	<i>/15</i>

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
				16-31	16-23	8-11	4-5
					24-31	12-15	6-7
			32-63	32-47	32-39	16-19	8-9
					40-47	20-23	10-11
				48-63	48-55	24-27	12-13
					56-63	28-31	14-15
			64-95	64-79	64-71	32-35	16-17
					72-79	36-39	18-19
				80-95	80-87	40-43	20-21
					88-95	44-47	22-23
				96-111	96-103	48-49	24-25
					104-111	50-51	26-27
			96-127	112-127	112-119	52-55	28-29
					120-127	56-59	30-31
			128-159	128-143	128-135	60-63	32-33
					136-143	64-67	34-35
				144-159	144-151	68-71	36-37
					152-159	72-75	38-39
			160-191	160-175	160-167	76-79	40-41
					168-175	80-83	42-43
				176-191	176-183	84-87	44-45
					184-191	88-89	46-47
			192-223	192-207	192-199	92-95	48-49
					200-207	96-99	50-51
				208-223	208-215	100-103	52-53
					216-223	104-107	54-55
			224-239	224-231	224-227	108-111	56-57
					232-239	112-115	58-59
			240-255	240-247	240-243	116-119	60-61
					248-255	120-123	62-63
					252-255	124-127	64-65
						128-131	66-67
						132-135	68-69
						136-139	70-71
						140-143	72-73
						144-147	74-75
						148-151	76-77
						152-155	78-79
						156-159	80-81
						160-163	82-83
						164-167	84-85
						168-171	86-87
						172-175	88-89
						176-179	90-91
						180-183	92-93
						184-187	94-95
						188-191	96-97
						192-195	98-99
						196-199	100-101
						200-203	102-103
						204-207	104-105
						208-211	106-107
						212-215	108-109
						216-219	110-111
						220-223	112-113
						224-227	114-115
						228-231	116-117
						232-235	118-119
						236-239	120-121
						240-243	122-123
						244-247	124-125
						248-251	126-127
						252-255	128-129
							130-131
							132-133
							134-135
							136-137
							138-139
							140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 2 of 4

The United States customers have a total of 8,388,608 addresses. Use the **Class A** address chart to break down the sub-subnetwork addresses for their different areas. At this stage of this problem you are creating sub-subnets in the second octet of the IP address.

Addresses Range: 75.0.0.0 to 75.127.255.255

Customer Name	Number of Addresses	Address Range	CIDR
Client #1	1,950,000	75.0.0.0 to 75.31.255.255	/11
Client #2	1,000,000	75.32.0.0 to 75.47.255.255	/12
Client #3	950,000	75.48.0.0 to 75.63.255.255	/12
Client #4	700,000	75.64.0.0 to 75.79.255.255	/12
Client #5	550,000	75.80.0.0 to 75.95.255.255	/12
Client #6	500,000	75.96.0.0 to 75.103.255.255	/13
Client #7	450,000	75.104.0.0 to 75.111.255.255	/13

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8	/9	/10	/11	/12	/13	/14	/15		
255.0.0.0 16,777,216 Hosts	255.128.0.0 8,388,608 Hosts	255.192.0.0 4,194,304 Hosts	255.224.0.0 2,097,152 Hosts	255.240.0.0 1,048,576 Hosts	255.248.0.0 524,288 Hosts	255.252.0.0 262,144 Hosts	255.254.0.0 131,072 Hosts		
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1		
						4-7	2-3		
				8-15		8-11	6-7		
						12-15	8-9		
				16-31	16-23	16-19	10-11		
					20-23	20-21	12-13		
				24-31	24-27	24-25	14-15		
					28-31	28-29	16-17		
				32-63	32-47	32-35	30-31	18-19	
						36-39	32-33	20-21	
		40-47	48-63	40-43	34-35	22-23			
				44-47	36-37	24-25			
		48-51	52-55	48-49	40-41	26-27			
				52-53	42-43	28-29			
		56-59	60-63	56-57	44-45	30-31			
				60-61	46-47	32-33			
		64-127	32-63	64-79	64-67	34-35	32-33		
					68-71	36-37	34-35		
				72-79	72-75	38-39	36-37		
					76-79	40-41	38-39		
				80-87	80-83	42-43	40-41		
					84-87	44-45	42-43		
				88-95	88-91	46-47	44-45		
					92-95	48-49	46-47		
			96-103	96-99	50-51	48-49			
				100-103	52-53	50-51			
			104-111	104-107	54-55	52-53			
				108-111	56-57	54-55			
			96-127	112-119	112-115	58-59	56-57		
					116-119	60-61	58-59		
				120-127	120-123	62-63	60-61		
					124-127	64-65	62-63		
				128-191	128-159	128-135	128-131	66-67	64-65
							132-135	68-69	66-67
			136-143			136-139	70-71	68-69	
						140-143	72-73	70-71	
		144-151	144-147			74-75	72-73		
			148-151			76-77	74-75		
		152-159	152-155			78-79	76-77		
			156-159			80-81	78-79		
		160-191	160-175		160-163	82-83	80-81		
					164-167	84-85	82-83		
			168-175		168-171	86-87	84-85		
					172-175	88-89	86-87		
			176-179		176-179	90-91	88-89		
					180-183	92-93	90-91		
			184-187		184-187	94-95	92-93		
					188-191	96-97	94-95		
		192-255	128-191	160-175	160-163	96-97	96-97		
					164-167	98-99	98-99		
				168-175	168-171	100-101	100-101		
					172-175	102-103	102-103		
				176-179	176-179	104-105	104-105		
					180-183	106-107	106-107		
				184-187	184-187	108-109	108-109		
					188-191	110-111	110-111		
			192-223	192-199	192-195	112-113	112-113		
					196-199	114-115	114-115		
				200-207	200-203	116-117	116-117		
					204-207	118-119	118-119		
				208-215	208-211	120-121	120-121		
					212-215	122-123	122-123		
				216-223	216-219	124-125	124-125		
					220-223	126-127	126-127		
	224-255	224-231	224-227	128-129	128-129				
			228-231	130-131	130-131				
		232-239	232-235	132-133	132-133				
			236-239	134-135	134-135				
		240-247	240-243	136-137	136-137				
			244-247	138-139	138-139				
		248-251	248-249	140-141	140-141				
			252-255	142-143	142-143				

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 3 of 4

Client #7 has a total of 524,288 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for their different clients. At this stage of this problem you are creating sub-subnets in the third or forth octet of the IP address.

Hint: Another way to look at this problem is to see that with the second octet range of 104 to 111 you have access to 8 groups of 65,536 addresses or 8 Class B VLSM charts.

ISP Addresses 75.104.0.0 to 75.111.255.255

Customer Name	Number of Addresses	Address Range	CIDR
Office #1	60,000	<i>75.104.0.0 to 75.104.255.255</i>	<i>/16</i>
Office #2	45,000	<i>75.105.0.0 to 75.105.255.255</i>	<i>/16</i>
Office #3	30,000	<i>75.106.0.0 to 75.106.127.255</i>	<i>/17</i>
Office #4	24,000	<i>75.106.128.0 to 75.106.255.255</i>	<i>/17</i>
Office #5	15,000	<i>75.107.0.0 to 75.107.63.255</i>	<i>/18</i>
Office #6	10,000	<i>75.107.64.0 to 75.107.127.255</i>	<i>/18</i>
Office #7	8,000	<i>75.107.128.0 to 75.107.159.255</i>	<i>/19</i>
Office #8	2,000	<i>75.107.160.0 to 75.107.167.255</i>	<i>/21</i>
Office #9	1,000	<i>75.107.168.0 to 75.107.171.255</i>	<i>/22</i>

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
							6-7
							8-9
							10-11
				16-31		12-15	12-13
						16-19	14-15
					16-23	20-23	16-17
							18-19
					24-31	24-27	20-21
						28-31	22-23
			32-63	32-47			24-25
							26-27
					32-39	32-35	28-29
						36-39	30-31
					40-47	40-43	32-33
							34-35
						44-47	36-37
							38-39
				48-63		48-51	40-41
					48-55	52-55	42-43
						56-59	44-45
					56-63	60-63	46-47
		64-127	64-95	64-79	64-71	64-67	48-49
						68-71	50-51
					72-79	72-75	52-53
						76-79	54-55
				80-95	80-87	84-87	56-57
					88-95	88-91	58-59
						92-95	60-61
						96-99	62-63
			96-127	96-103		100-103	64-65
						104-107	66-67
					104-111	108-111	68-69
							70-71
				112-127		112-115	72-73
					112-119	116-119	74-75
							76-77
					120-127	124-127	78-79
		128-191	128-143	128-143	128-135	128-131	80-81
						132-135	82-83
					136-143	140-143	84-85
							86-87
			144-159	144-151	144-147	148-151	88-89
						152-155	90-91
					152-159	156-159	92-93
							94-95
			160-191	160-175	160-167	164-167	96-97
						168-171	98-99
					168-175	172-175	100-101
							102-103
			176-191	176-183	176-179	180-183	104-105
						184-187	106-107
					184-191	188-191	108-109
							110-111
		192-255	192-223	192-207	192-199	192-195	112-113
						196-199	114-115
					200-207	204-207	116-117
							118-119
			208-223	208-215		212-215	120-121
						216-219	122-123
					216-223	220-223	124-125
							126-127
			224-239	224-231	224-231	228-231	128-129
						232-235	130-131
					232-239	236-239	132-133
							134-135
			240-255	240-247	240-243	244-247	136-137
						248-251	138-139
					248-255	252-255	140-141
							142-143
							144-145
							146-147
							148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

VLSM

with

Class A and B Addresses

Sample Problem 38

Part 4 of 4

Office #7 from part 3 of 4 has a total of 8,192 addresses. Use the **Class B** address chart to break down the sub-subnetwork addresses for the different branch offices. At this stage of this problem you are creating sub-subnets in the third octet of the IP address.

Hint: Remember that the range of this problem is between 128 and 159 in the third octet. Your subnetting will start in the middle of the chart not at the top for this range.

ISP Addresses 75.107.128.0 to 75.107.159.255

Customer Name	Number of Addresses	Address Range	CIDR
Branch #1	4,000	75.107.128.0 to 75.107.143.255	/20
Branch #2	2,000	75.107.144.0 to 75.107.151.255	/21
Branch #3	1,000	75.107.152.0 to 75.107.155.255	/22
Branch #4	500	75.107.156.0 to 75.107.159.255	/23
Branch #5	450	75.107.160.0 to 75.107.161.255	/23

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
			16-31	16-31	16-23	16-19	8-9
						20-23	10-11
					24-31	24-27	12-13
						28-31	14-15
			32-63	32-47	32-39	32-35	16-17
						36-39	18-19
					40-47	40-43	20-21
						44-47	22-23
				48-63	48-55	48-51	24-25
						52-55	26-27
					56-63	56-59	28-29
						60-63	30-31
		64-127	64-95	64-79	64-71	64-67	32-33
						68-71	34-35
					72-79	72-75	36-37
						76-79	38-39
			80-95	80-87	80-83	80-81	40-41
						84-87	42-43
					88-95	88-91	44-45
						92-95	46-47
			96-127	96-103	96-99	96-97	48-49
						100-103	50-51
					104-111	104-107	52-53
						108-111	54-55
				112-127	112-119	112-115	56-57
						116-119	58-59
					120-127	120-123	60-61
						124-127	62-63
	128-255	128-191	See Hint 128-159	128-143	128-135	128-131	64-65
						132-135	66-67
					136-143	136-139	68-69
						140-143	70-71
			144-159	144-151	144-147	144-145	72-73
						148-151	74-75
					152-159	152-155	76-77
						156-159	78-79
			160-191	160-175	160-167	160-163	80-81
						164-167	82-83
					168-175	168-171	84-85
						172-175	86-87
				176-191	176-183	176-179	88-89
						180-183	90-91
					184-191	184-187	92-93
						188-191	94-95
		192-255	192-223	192-207	192-199	192-195	96-97
						196-199	98-99
					200-207	200-203	100-101
						204-207	102-103
				208-223	208-215	208-211	104-105
						212-215	106-107
					216-223	216-219	108-109
						220-223	110-111
			224-255	224-239	224-231	224-227	112-113
						228-231	114-115
					232-239	232-235	116-117
						236-239	118-119
				240-255	240-247	240-243	120-121
						244-247	122-123
					248-255	248-251	124-125
						252-255	126-127

Reference Charts and Support Materials

Class A Addresses
VLSM Chart 8-15 Bits (2nd octet)

[illegible]

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

16 16-20 20,000 seeds	17 17-21 21,700 seeds	18 18-22 22,500 seeds	19 19-23 23,100 seeds	20 20-24 24,000 seeds	21 21-25 25,000 seeds	22 22-26 26,000 seeds	23 23-27 27,000 seeds
			9-81	3-15	6-15	2-15	1-15
				13-21	5-21	2-21	1-21
				19-21	10-21	5-21	3-21
			9-81	50-47	30-49	10-49	5-49
				41-47	21-47	10-47	5-47
			9-81	40-55	40-55	30-55	20-55
				50-55	50-55	40-55	30-55
				60-71	60-71	50-71	40-71
				71-75	71-75	60-75	50-75
			9-81	80-87	80-87	70-87	60-87
				88-95	88-95	78-95	68-95
				96-111	96-111	86-111	76-111
			9-81	912-107	152-109	102-109	52-109
				108-127	108-127	98-127	88-127
				128-145	128-145	118-145	108-145
				146-161	146-161	136-161	126-161
				162-192	162-192	152-192	142-192
				193-198	193-198	183-198	173-198
				199-175	199-175	189-175	179-175
			180-191	176-191	176-191	166-191	156-191
				176-191	176-191	166-191	156-191
				182-207	182-207	172-207	162-207
				208-225	208-225	198-225	188-225
				226-228	226-228	216-228	206-228
				229-239	229-239	219-239	209-239
			224-265	240-255	240-255	230-255	220-255
				256-255	256-255	246-255	236-255

Class C Addresses
VLSM Chart 24-30 Bits (4th octet)

[illegible]

Class A Addresses

VLSM Chart 8-15 Bits (2nd octet)

/8 255.0.0.0 16,777,216 Hosts	/9 255.128.0.0 8,388,608 Hosts	/10 255.192.0.0 4,194,304 Hosts	/11 255.224.0.0 2,097,152 Hosts	/12 255.240.0.0 1,048,576 Hosts	/13 255.248.0.0 524,288 Hosts	/14 255.252.0.0 262,144 Hosts	/15 255.254.0.0 131,072 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
						4-7	2-3
					8-15	8-11	4-5
						12-15	6-7
							8-9
							10-11
							12-13
							14-15
			16-31		16-23	16-19	16-17
						20-23	18-19
					24-31	24-27	20-21
						28-31	22-23
						32-35	24-25
						36-39	26-27
						40-43	28-29
						44-47	30-31
						48-51	32-33
						52-55	34-35
						56-59	36-37
						60-63	38-39
			32-63	32-47		64-67	40-41
					40-47	68-71	42-43
						72-75	44-45
						76-79	46-47
				48-63	48-55	80-83	48-49
						84-87	50-51
						88-91	52-53
						92-95	54-55
			64-127	64-79	64-71	96-99	56-57
						100-103	58-59
					80-87	104-107	60-61
						108-111	62-63
				80-95	80-87	112-115	64-65
						116-119	66-67
					88-95	120-123	68-69
						124-127	70-71
				96-111	96-103	128-131	72-73
						132-135	74-75
					104-111	136-139	76-77
						140-143	78-79
				96-127	112-119	144-147	80-81
						148-151	82-83
					120-127	152-155	84-85
						156-159	86-87
	128-255	128-191	128-159	128-143	128-135	160-167	88-89
						164-167	90-91
					136-143	168-171	92-93
						172-175	94-95
			144-159		144-151	176-179	96-97
						180-183	98-99
					152-159	184-187	100-101
						188-191	102-103
			160-191	160-175	160-167	192-195	104-105
						196-199	106-107
					168-175	200-203	108-109
						204-207	110-111
				176-191	176-183	208-211	112-113
						212-215	114-115
					184-191	216-219	116-117
						220-223	118-119
			192-255	192-207	192-199	224-227	120-121
						228-231	122-123
					200-207	232-235	124-125
						236-239	126-127
				208-223	208-215	240-243	128-129
						244-247	130-131
					216-223	248-251	132-133
						252-255	134-135
				224-239	224-231		136-137
							138-139
					232-239		140-141
							142-143
				240-255	240-247		144-145
							146-147
					248-255		148-149
							150-151
							152-153
							154-155
							156-157
							158-159
							160-161
							162-163
							164-165
							166-167
							168-169
							170-171
							172-173
							174-175
							176-177
							178-179
							180-181
							182-183
							184-185
							186-187
							188-189
							190-191
							192-193
							194-195
							196-197
							198-199
							200-201
							202-203
							204-205
							206-207
							208-209
							210-211
							212-213
							214-215
							216-217
							218-219
							220-221
							222-223
							224-225
							226-227
							228-229
							230-231
							232-233
							234-235
							236-237
							238-239
							240-241
							242-243
							244-245
							246-247
							248-249
							250-251
							252-253
							254-255

Class B Addresses

VLSM Chart 16-23 Bits (3rd octet)

/16 255.255.0.0 65,536 Hosts	/17 255.255.128.0 32,768 Hosts	/18 255.255.192.0 16,384 Hosts	/19 255.255.224.0 8,192 Hosts	/20 255.255.240.0 4,096 Hosts	/21 255.255.248.0 2,048 Hosts	/22 255.255.252.0 1,024 Hosts	/23 255.255.254.0 512 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3	0-1
					8-15	4-7	2-3
						8-11	4-5
						12-15	6-7
						16-19	8-9
						20-23	10-11
			16-31	16-31	16-23	16-19	12-13
					24-31	20-23	14-15
						24-27	16-17
						28-31	18-19
						32-35	20-21
						36-39	22-23
			32-63	32-47	32-39	32-35	24-25
					40-47	40-43	26-27
						44-47	28-29
						48-51	30-31
						52-55	32-33
						56-59	34-35
			48-63	48-55	56-63	56-59	36-37
						60-63	38-39
						64-67	40-41
						68-71	42-43
						72-75	44-45
						76-79	46-47
		64-127	64-95	64-79	64-71	64-67	48-49
					72-79	68-71	50-51
						72-75	52-53
						80-83	54-55
						84-87	56-57
						88-91	58-59
			80-95	80-87	88-95	88-91	60-61
						92-95	62-63
						96-99	64-65
						100-103	66-67
						104-107	68-69
						108-111	70-71
			96-127	96-111	112-119	112-115	72-73
					120-127	116-119	74-75
						124-127	76-77
						128-131	78-79
						132-135	80-81
						136-139	82-83
	128-255	128-191	128-159	128-143	128-135	128-131	84-85
					136-143	132-135	86-87
						140-143	88-89
						144-147	90-91
						148-151	92-93
						152-155	94-95
			144-159	144-151	152-155	152-155	96-97
						156-159	98-99
						160-163	100-101
						164-167	102-103
						168-171	104-105
						172-175	106-107
			160-191	160-175	176-179	176-179	108-109
					180-183	180-183	110-111
						184-187	112-113
						188-191	114-115
						192-195	116-117
						196-199	118-119
		192-255	192-223	192-207	200-207	200-203	120-121
						204-207	122-123
						208-211	124-125
						212-215	126-127
						216-219	128-129
						220-223	130-131
			224-255	224-231	224-227	224-227	132-133
					228-231	228-231	134-135
					232-235	232-235	136-137
					236-239	236-239	138-139
					240-243	240-243	140-141
					244-247	244-247	142-143
			240-255	240-247	248-251	248-251	144-145
					252-255	252-255	146-147
							148-149
							150-151
							152-153
							154-155

Class C Addresses

VLSM Chart 24-30 Bits (4th octet)

/24	/25	/26	/27	/28	/29	/30
255.255.255.0 256 Hosts	255.255.255.128 128 Hosts	255.255.255.192 64 Hosts	255.255.255.224 32 Hosts	255.255.255.240 16 Hosts	255.255.255.248 8 Hosts	255.255.255.252 4 Hosts
0 - 255	0-127	0-63	0-31	0-15	0-7	0-3
						4-7
					8-15	8-11
						12-15
				16-31	16-23	16-19
						20-23
					24-31	24-27
						28-31
			32-63	32-47	32-39	32-35
						36-39
					40-47	40-43
						44-47
				48-63	48-55	48-51
						52-55
					56-63	56-59
						60-63
		64-127	64-95	64-79	64-71	64-67
						68-71
					72-79	72-75
						76-79
				80-95	80-87	80-83
						84-87
					88-95	88-91
						92-95
			96-127	96-111	96-103	96-99
						100-103
					104-111	104-107
						108-111
				112-127	112-119	112-115
						116-119
					120-127	120-123
						124-127
	128-255	128-191	128-159	128-143	128-135	128-131
						132-135
					136-143	136-139
						140-143
				144-159	144-151	144-147
						148-151
					152-159	152-155
						156-159
			160-191	160-175	160-167	160-163
						164-167
					168-175	168-171
						172-175
				176-191	176-183	176-179
						180-183
					184-191	184-187
						188-191
		192-255	192-223	192-207	192-199	192-195
						196-199
					200-207	200-203
						204-207
				208-223	208-215	208-211
						212-215
					216-223	216-219
						220-223
			224-255	224-239	224-231	224-227
						228-231
					232-239	232-235
						236-239
				240-255	240-247	240-243
						244-247
					248-255	248-251
						252-255

Class A Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/8	0	255.0.0.0	1	16,777,216	16,777,214
/9	1	255.128.0.0	2	8,388,608	8,388,606
/10	2	255.192.0.0	4	4,194,304	4,194,302
/11	3	255.224.0.0	8	2,097,152	2,097,150
/12	4	255.240.0.0	16	1,048,576	1,048,574
/13	5	255.248.0.0	32	524,288	524,286
/14	6	255.252.0.0	64	262,144	262,142
/15	7	255.254.0.0	128	131,072	131,070
/16	8	255.255.0.0	256	65,536	65,534
/17	9	255.255.128.0	512	32,768	32,766
/18	10	255.255.192.0	1,024	16,384	16,382
/19	11	255.255.224.0	2,048	8,192	8,190
/20	12	255.255.240.0	4,096	4,096	4,094
/21	13	255.255.248.0	8,192	2,048	2,046
/22	14	255.255.252.0	16,384	1,024	1,022
/23	15	255.255.254.0	32,768	512	510
/24	16	255.255.255.0	65,536	256	254
/25	17	255.255.255.128	131,072	128	126
/26	18	255.255.255.192	262,144	64	62
/27	19	255.255.255.224	524,288	32	30
/28	20	255.255.255.240	1,048,576	16	14
/29	21	255.255.255.248	2,097,152	8	6
/30	22	255.255.255.252	4,194,304	4	2

Class B Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/16	0	255.255.0.0	1	65,536	65,534
/17	1	255.255.128.0	2	32,768	32,766
/18	2	255.255.192.0	4	16,384	16,382
/19	3	255.255.224.0	8	8,192	8,190
/20	4	255.255.240.0	16	4,096	4,094
/21	5	255.255.248.0	32	2,048	2,046
/22	6	255.255.252.0	64	1,024	1,022
/23	7	255.255.254.0	128	512	510
/24	8	255.255.255.0	256	256	254
/25	9	255.255.255.128	512	128	126
/26	10	255.255.255.192	1,024	64	62
/27	11	255.255.255.224	2,048	32	30
/28	12	255.255.255.240	4,096	16	14
/29	13	255.255.255.248	8,192	8	6
/30	14	255.255.255.252	16,384	4	2

Class C Addressing Guide					
CIDR	# of Bits Borrowed	Subnet Mask	Total # of Subnets	Total # of Hosts	Usable # of Hosts
/24	0	255.255.255.0	1	256	254
/25	1	255.255.255.128	2	128	126
/26	2	255.255.255.192	4	64	62
/27	3	255.255.255.224	8	32	30
/28	4	255.255.255.240	16	16	14
/29	5	255.255.255.248	32	8	6
/30	6	255.255.255.252	64	4	2